

Locally Recoverable Data Availability Sampling

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Abstract. Data Availability Sampling (DAS) enables light clients to probabilistically verify that a large published object remains retrievable by sampling a small number of coded symbols and checking consistency. Most existing DAS designs are predicated on MDS-style global recovery thresholds and therefore yield an essentially binary availability signal: either sufficient evidence accumulates to enable global reconstruction, or it does not. In this work we introduce *Locally Recoverable Data Availability Sampling* (LR-DAS), which leverages locally recoverable codes with *disjoint* recovery groups to provide *graded* availability evidence. A verifier sequentially certifies groups: once it has verified r distinct openings in a group, it can reconstruct the entire group locally and monotonically increase a certified availability fraction.

The central cryptographic challenge is preventing *splicing* attacks, where an adversary mixes locally consistent views that do not arise from a single global codeword. We define *locality-aware erasure-code commitments* that formalize position binding, local binding, and an explicit local-global consistency requirement, and we give a construction that enforces local-global consistency using a single out-of-domain algebraic linking check per certified group. We prove graded soundness via overshoot bounds that decompose into a sampling term and a negligible cryptographic failure term, covering both missing-groups and missing-fraction withholding patterns. We provide two concrete instantiations: a succinct pairing-based realization and a transparent realization based on low-degree/proximity testing, and we evaluate the resulting trade-offs against representative DAS and repair-oriented approaches.

Keywords: Data Availability Sampling · Locally Recoverable Codes · Tamo-Barg Codes · Erasure-Code Commitment (KZG-based, FRI based) · Graded Availability

1 Introduction

Data availability and light verification. Many distributed systems publish large data objects that must remain retrievable after publication. A prominent example is blockchain scalability, where “light” verifiers should be able to decide

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whether block data is available without downloading the full block. *Data availability sampling* (DAS) addresses this goal by committing to an erasure-coded representation of the data and then allowing verifiers to query a small number of randomly chosen positions; if the queried positions can be opened consistently, the verifier gains probabilistic confidence that the underlying data is available. A systematic treatment of DAS and erasure-code commitments is given in [7], and recent constructions instantiate this framework efficiently using polynomial commitments and IOPP techniques, e.g., FRIDA [8].

A limitation of MDS-based DAS: binary semantics. Most deployed and proposed DAS designs rely on maximum-distance separable (MDS) codes, most notably Reed–Solomon (RS) codes. MDS codes are ideal for *global* recovery: once enough symbols are known, the entire message is uniquely determined. However, this global threshold also induces an inherently *binary* verification semantics for light clients: until the verifier has sampled enough positions to meet a global recovery condition, the verifier cannot certify meaningful partial progress toward availability. Two-dimensional RS layouts (rows/columns) and their variants improve operational aspects (e.g., organization and repair paths), but the fundamental “all-or-nothing” availability signal remains tied to a global decoding threshold [1,6].

From binary to graded availability via locality. We demonstrate that, in many distributed settings, offering *graded* availability provides distinct advantages to a light verifier: instead of returning a single bit that transitions only near a global threshold, the verifier should be able to certify *how much* of the data has been verified as available. Graded signals can support earlier decision-making (e.g., early relay policies), faster anomaly detection (identifying which parts remain unavailable), and targeted re-sampling or repair efforts.

We achieve graded availability by encoding the data using a *locally recoverable code* (LRC). Concretely, we partition the codeword positions into ℓ disjoint *local groups* $A_0, \dots, A_{\ell-1}$, each of size $r + \alpha$. Each group behaves as a local $(r + \alpha, r)$ MDS code: *any* r symbols in A_j uniquely determine the remaining α symbols in A_j . Thus, once a verifier has obtained and verified r distinct openings from a group A_j , the verifier can interpolate the unique degree- $< r$ local polynomial δ_j and treat the entire group as *certified available*. We measure progress by an *availability score* that counts how many groups have been certified from the verifier’s sampled-and-verified set of openings, and we output the corresponding certified fraction in $[0, 1]$. To obtain strong global distance while preserving this locality structure, we instantiate the coding layer with optimal Tamo–Barg LRCs [13].

A cryptographic obstacle: preventing cross-group splicing. Local certification alone is not sufficient. An adversary could attempt to “splice” together locally consistent groups that do *not* arise as restrictions of any single global codeword, thereby deceiving a verifier that checks only local constraints. To rule out such attacks, we define and construct a *locality-aware erasure-code commitment* that

enforces a precise *local-global consistency* property: every accepted local opening must be consistent with a *single* globally committed codeword (formalized later as a binding/extractability-style notion in the style of [7]). Our key technique is a single algebraic “link” between a global univariate polynomial and each local polynomial, verified at a fresh out-of-domain point. This link is compatible with standard polynomial commitment schemes such as KZG [10] and transparent FRI-based stacks [3,8].

Our approach: Locally Recoverable Data Availability Sampling (LR-DAS). We introduce *Locally Recoverable DAS* (LR-DAS), a DAS framework whose verification unit is a local recovery group. At a high level, the committer encodes a message into a Tamo–Barg codeword and commits to a global univariate polynomial whose evaluations define the codeword. To certify a group A_j , a light verifier requests openings of positions in A_j until it has verified r distinct points; it then reconstructs the full group locally and increases its availability score. A single local–global linking check prevents splicing and ensures that all accepted local certificates assemble into one global codeword.

Comparison with prior work. Our design is complementary to prior approaches. Two-dimensional RS constructions and Ethereum-inspired designs organize the codeword geometrically and can offer multiple repair paths, but they retain binary availability semantics tied to global decoding thresholds [1,6,11,12]. Multi-DAS [5] improves *repair* using multiplicity-style codes and multivariate/derivative-aware commitments; LR-DAS instead focuses on *graded availability semantics* via LRC locality and keeps a univariate pipeline with a simple linking mechanism. Our goal is not to replace these designs universally, but to provide a robust primitive and analysis for scenarios where progressive, group-level certification is valuable.

Contributions.

- **Graded availability for DAS from LRC locality.** We formalize a graded verification objective for DAS in which the verifier outputs a certified availability fraction based on how many local groups have been certified, rather than a single threshold bit.
- **Locality-aware erasure-code commitments with local–global consistency.** We define a commitment interface and security notions that make splicing attacks explicit and rule them out by requiring that all accepted local openings are consistent with a single globally committed codeword, building on the erasure-code commitment framework of [7].
- **A one-link construction for Tamo–Barg codes.** We give a locality-aware commitment structure in which each accepted local certificate is tied to the global commitment via one algebraic out-of-domain check, enabling succinct verification.
- **Two instantiations: succinct and transparent.** We instantiate LR-DAS using (i) KZG-based polynomial commitments [10] and (ii) transpar-

ent FRI/IOPP techniques [3,8], and we show how batching and aggregation techniques for polynomial openings apply in our setting.

- **Graded soundness bounds.** We prove overshoot bounds for natural withholding models (missing-fraction and missing-groups), showing that false certification factors into a sampling term and a negligible cryptographic failure term.

Scope, parameters, and practical considerations. LR-DAS exposes a design space controlled by (r, α) : smaller r improves responsiveness (fewer verified openings per group), while larger α improves tolerance to missing symbols within a group and typically improves global distance at the cost of larger groups. Graded certification is most informative when losses are diffuse across many groups; concentrated burst losses may require increasing α , adding global redundancy, or adjusting sampling to ensure coverage across groups. Finally, our local-global link is a Schwartz–Zippel-style out-of-domain check; its soundness can be amplified by repeating the link at multiple points or by working over an appropriate field/extension, and we make these trade-offs explicit in the parameter discussion.

2 Preliminaries and Notation

This section fixes notation and recalls the coding- and commitment-theoretic primitives used throughout. We provide formal definitions for all objects prior to their use to ensure notational consistency (e.g., we use a single notation $Z_S(X)$ for vanishing polynomials for *every* set S).

2.1 Basic notation

Let $\mathbb{F}_q$ be a finite field of size q . For a polynomial $f(X) \in \mathbb{F}_q[X]$, we write $\deg(f)$ for its degree. For a finite set $S \subseteq \mathbb{F}_q$, we define the *vanishing polynomial*

$$Z_S(X) := \prod_{a \in S} (X - a) \in \mathbb{F}_q[X]. \quad (1)$$

We will use $Z_S(X)$ consistently for all sets (including local groups and query sets). For a subset $U \subseteq S$, we write $f(X) \bmod Z_U(X)$ for the unique remainder polynomial of degree $< |U|$.

Evaluation domain and locality partition. We fix an evaluation set $A \subseteq \mathbb{F}_q$ of size $|A| = n$, equipped with a partition into ℓ *disjoint local groups*

$$A = \bigsqcup_{j=0}^{\ell-1} A_j, \quad |A_j| = r + \alpha, \quad (2)$$

for integers $r \geq 1$ and $\alpha \geq 1$. Thus $n = \ell(r + \alpha)$. Intuitively, r represents the number of symbols sufficient to reconstruct a group, and α is the local erasure tolerance.

Table 1. Notation used throughout the paper.

Symbol	Meaning
$\mathbb{F}_q$	Finite field of size q .
$A \subseteq \mathbb{F}_q$	Evaluation domain, $ A = n$.
$A_0, \dots, A_{\ell-1}$	Disjoint local groups, $A = \bigsqcup_j A_j$, $ A_j = r + \alpha$.
r, α, ℓ, n, k, d	Locality parameter, local tolerance, number of groups, code length, dimension, distance.
$Z_S(X)$	Vanishing polynomial $\prod_{a \in S} (X - a)$ for any finite $S \subseteq \mathbb{F}_q$.
$f(X)$	A global polynomial whose evaluations on A define a codeword.
$\delta_j(X)$	A local polynomial of degree $< r$ whose evaluations on A_j define the j -th local group.

A standard polynomial identity test. We will repeatedly use the fact that for any nonzero polynomial $p(X) \in \mathbb{F}_q[X]$ of degree at most t , a uniformly random $\rho \leftarrow \mathbb{F}_q$ satisfies $p(\rho) = 0$ with probability at most t/q (Schwartz–Zippel). We will also use out-of-domain points $\rho \in \mathbb{F}_q \setminus A$.

2.2 Locally recoverable codes with disjoint recovery groups

A (block) code $C \subseteq \mathbb{F}_q^n$ has *disjoint $(r + \alpha, r)$ recovery groups* with respect to the partition (2) if for every codeword $c \in C$ and every group index j there exists a polynomial $\delta_j(X) \in \mathbb{F}_q[X]$ of degree $< r$ such that

$$c|_{A_j} = (\delta_j(a))_{a \in A_j}. \quad (3)$$

Equivalently, each punctured code $C|_{A_j}$ is an $(r + \alpha, r)$ MDS code. Consequently, *any* set of r distinct symbols in A_j uniquely determines the remaining α symbols in A_j via interpolation of δ_j . This is the locality structure exploited by LR-DAS: a verifier that has verified r distinct openings in group A_j can treat the entire group as locally reconstructed.

Distance and design trade-offs. For codes with all-symbol locality r and group size $r + \alpha$, the minimum distance is upper bounded by a Singleton-type bound, and Tamo–Barg codes achieve this bound with equality [13]. In LR-DAS, (r, α) are *design parameters*: smaller r yields finer-grained certification (fewer openings needed per group), while larger α increases local erasure tolerance but enlarges groups (more payload per certified group).

Taxonomy of Reconstruction Thresholds. We will later distinguish carefully between: (i) information-theoretic message dimension k (there exist information sets of size k), (ii) worst-case erasure recovery thresholds tied to distance d (e.g., $n - d + 1$ known symbols suffice for unique decoding), and (iii) polynomial interpolation thresholds tied to a global degree bound D (e.g., $D + 1$ points fix a degree- $\leq D$ polynomial). Keeping these notions separate avoids confusion when analyzing random sampling versus adversarial erasures.

2.3 Tamo–Barg codes as univariate LRC evaluation codes

We use optimal Tamo–Barg (TB) locally recoverable codes [13]. TB codes can be described as univariate evaluation codes with a built-in locality partition.

Good polynomial and group labels. Fix $A = \bigsqcup_{j=0}^{\ell-1} A_j \subseteq \mathbb{F}_q$ with $|A_j| = r + \alpha$. A polynomial $g(X) \in \mathbb{F}_q[X]$ is *good for the partition* if it is constant on each group: there exist pairwise distinct labels $\gamma_0, \dots, \gamma_{\ell-1} \in \mathbb{F}_q$ such that

$$g(a) = \gamma_j \quad \text{for all } a \in A_j \text{ and all } j \in \{0, \dots, \ell-1\}. \quad (4)$$

(Concrete constructions of such g are given in [13].)

Encoding polynomial and local restrictions. Let $k \leq \ell r$ be the message length. Write $k = \sum_{i=0}^{r-1} \kappa_i$ where each $\kappa_i \in \{\lfloor k/r \rfloor, \lceil k/r \rceil\}$. A TB encoding can be described by choosing coefficient polynomials $f_i(X) \in \mathbb{F}_q[X]$ of the form

$$f_i(X) = \sum_{t=0}^{\kappa_i-1} a_{i,t} g(X)^t, \quad i = 0, \dots, r-1, \quad (5)$$

for message coefficients $a_{i,t} \in \mathbb{F}_q$, and then assembling the global polynomial

$$f(X) = \sum_{i=0}^{r-1} f_i(X) X^i. \quad (6)$$

The codeword is the evaluation vector $c = (f(a))_{a \in A} \in \mathbb{F}_q^n$.

Because g is constant on each group A_j , each coefficient $f_i(X)$ restricts to a constant on A_j , so f restricts to a degree- $< r$ polynomial on A_j . Concretely, for each j there exists $\delta_j(X) \in \mathbb{F}_q[X]$ with $\deg(\delta_j) < r$ such that

$$(f(a))_{a \in A_j} = (\delta_j(a))_{a \in A_j}. \quad (7)$$

Thus TB codes satisfy the disjoint recovery-group property of Section 2.2.

Global degree bound. Let D be an upper bound on $\deg(f)$ under the above construction. A standard bound is

$$\deg(f) \leq D := \max_{0 \leq i \leq r-1} ((\kappa_i - 1) \deg(g) + i), \quad (8)$$

and for the canonical TB choices one has $\deg(g) = r + \alpha$. We will use D later when stating global interpolation thresholds and when parameterizing polynomial commitments.

Remark: TB Codes as Structured Subcodes. TB codewords form a structured subset of RS evaluations: enforcing that each restriction (7) has degree $< r$ is *not* sufficient, by itself, to characterize TB membership. Our protocols therefore maintain both (i) local RS-type constraints (degree $< r$ on each group) and (ii) a global commitment to a *single* polynomial f of degree $\leq D$ that is consistent with every accepted local view. Later sections formalize the corresponding local–global consistency requirement and show how it prevents splicing attacks.

2.4 Abstraction of Polynomial Commitments

We treat polynomial commitment schemes (PCS) as abstract primitives. A PCS for degree bound D consists of algorithms (**Setup**, **Commit**, **Open**, **Verify**) where **Verify** accepts only if $y = f(x)$. We rely primarily on *evaluation binding* (no adversary can open the same commitment to two different values at the same point), and on standard extensions that support batching/multi-point openings.

KZG. KZG commitments provide succinct (constant-size) openings in pairing groups under a structured reference string [10].

FRI/IOPP. FRI provides a transparent interactive oracle proof of proximity for RS codes and serves as a foundation for transparent PCS/commitment stacks [3]. In our transparent instantiation, Merkle commitments provide position binding and FRI-style tests provide low-degree/proximity guarantees.

Multi-point openings via vanishing polynomials (optional tool). When we need to open a polynomial at many points in a set U , we will use the standard “remainder/quotient” method based on $Z_U(X)$ [4]: send the claimed evaluations (equivalently, the remainder polynomial of degree $< |U|$), and prove that $f(X) = R_U(X) + Z_U(X) \cdot W_U(X)$ for some quotient $W_U(X)$.

3 LR-DAS: Model and Syntax

This section formalizes LR-DAS as a DAS wrapper around (i) an LRC encoding with disjoint recovery groups and (ii) a locality-aware erasure-code commitment. We focus on *syntax* and *certification semantics* here; security definitions and proofs appear later. Our model and terminology follow the erasure-code commitment viewpoint of [7].

3.1 Participants, public parameters, and execution

Public parameters. Fix a finite field $\mathbb{F}_q$, an evaluation domain $A \subseteq \mathbb{F}_q$ with $|A| = n$, and a partition into ℓ disjoint local groups $A = \bigsqcup_{j=0}^{\ell-1} A_j$ with $|A_j| = r + \alpha$ (Section 2.1). We work with an (n, k, d) code $C \subseteq \mathbb{F}_q^n$ that is locally recoverable with respect to this partition (Section 2.2); in particular, each restriction $C|_{A_j}$ is an $(r + \alpha, r)$ MDS code. Our main instantiation uses Tamo–Barg codes (Section 2.3), but the LR-DAS syntax below applies to any code with this locality structure.

Roles. A *committer* (also called the prover) encodes a message and publishes a commitment to the resulting codeword. A *verifier* (light client) samples a small set of positions and requests openings. Based on accepted openings, the verifier outputs a *graded availability certificate*—a number in $[0, 1]$ reflecting how many local groups have been certified.

One execution of LR-DAS (verifier view).

1. **Commit:** The committer encodes a message m into a codeword $c \in C$ and publishes a commitment com to c (and keeps internal state St).
2. **Sample:** The verifier derives a seed after seeing com and samples a multiset Q of positions in A according to a policy (uniform or group-aware).
3. **Open:** For each group index j , let $U_j := \text{supp}(Q \cap A_j)$ be the set of *distinct* sampled points in group A_j . The committer returns an opening proof π_j that reveals the values $\{c(a)\}_{a \in U_j}$.
4. **Verify:** The verifier checks each opening. Accepted openings are accumulated into a set $S \subseteq A \times \mathbb{F}_q$ (i.e., a set containing pairs of the coordinate (or index) and the symbol itself, e.g., (i, c_i)).
5. **Certify:** A group A_j is *certified* if S contains at least r accepted, distinct points from A_j and these points are consistent with a degree- $< r$ local view that is also consistent with the committed global codeword. The verifier outputs a certified fraction $\phi \in [0, 1]$ based on how many groups are certified.

Fig. 1. LR-DAS interaction. The opening/verification interface is provided by a locality-aware erasure-code commitment (formalized later).

Randomness and non-adaptive withholding. We analyze the standard non-adaptive withholding model: the committer fixes which symbols to withhold *before* the verifier’s sampling randomness is determined. In practice, the verifier derives sampling randomness *after* seeing the commitment, using a domain-separated Fiat–Shamir seed or an external randomness beacon. This prevents the committer from tailoring withheld positions to unseen queries. (Adaptive withholding and grinding considerations can be addressed with standard mitigations; we keep the core model simple here and quantify cryptographic failure probabilities later.)

3.2 LR-DAS syntax

LR-DAS is parameterized by a locality-aware commitment scheme that can open *subsets within a local group* and bind them to a *single* global codeword. This is exactly the kind of erasure-code commitment interface advocated in [7]. At this point we only need the interface (syntax), not its security definitions.

Syntax of Locality-Aware Commitments. We assume a commitment scheme with algorithms

$$\begin{aligned}
&\text{Setup}(1^\lambda) \rightarrow \text{ck}, \\
&\text{Com}(\text{ck}, m) \rightarrow (\text{com}, \text{St}), \\
&\text{Open}(\text{ck}, \text{St}, j, U) \rightarrow \pi, \\
&\text{Verify}(\text{ck}, \text{com}, j, U, \pi) \in \{0, 1\}
\end{aligned}$$

where $j \in \{0, \dots, \ell - 1\}$ and $U \subseteq A_j$.

An opening proof π reveals the values $\{c(a)\}_{a \in U}$ and convinces the verifier that these values are valid for the committed codeword. Later sections specify the exact verification conditions needed for LR-DAS (local degree- $< r$ membership and local-global consistency).

Sampling policy. A sampling policy is any algorithm that, given $(A, \{A_j\}, \text{com})$ and a sampling budget s , outputs a multiset Q of s sampled positions in A . We will consider: (i) *uniform sampling* over A , and (ii) *group-aware sampling* that assigns per-group budgets $s_0, \dots, s_{\ell-1}$ with $\sum_j s_j = s$ and samples s_j positions from each A_j . In both cases, repeated draws are allowed, and the LR-DAS certificate depends on the number of *distinct* sampled points per group.

3.3 Certified groups and graded availability

Accepted openings. Let Q be the multiset of sampled positions and let $U_j = \text{supp}(Q \cap A_j)$ be the distinct sampled set in group A_j . For each j , the verifier requests $\pi_j \leftarrow \text{Open}(\text{ck}, \text{St}, j, U_j)$ and accepts it iff $\text{Verify}(\text{ck}, \text{com}, j, U_j, \pi_j) = 1$. Let S denote the set of accepted pairs (a, v) , where $a \in A$ and $v \in \mathbb{F}_q$ is the opened value.

Locally certified group. We say that a group A_j is *locally certified from S* if both conditions hold:

1. (*enough distinct points*) S contains at least r distinct accepted points from A_j ; and
2. (*local structure and global consistency*) the accepted values in A_j are consistent with a single degree- $< r$ polynomial δ_j that is also consistent with the same globally committed codeword.

When A_j is locally certified, the verifier can interpolate δ_j from any r accepted points and reconstruct all $r + \alpha$ symbols in A_j by evaluating δ_j on A_j .

Availability score and certified fraction. Let $\mathbf{1}\{\cdot\}$ denote the indicator function. Define the (monotone) score

$$A(S) := \sum_{j=0}^{\ell-1} \mathbf{1}\{A_j \text{ is locally certified from } S\}, \quad (9)$$

and the corresponding certified availability fraction

$$\phi(S) := \frac{A(S)}{\ell} \in [0, 1]. \quad (10)$$

LR-DAS outputs $\phi(S)$, i.e., the fraction of local groups that have been certified. This is the graded analogue of the binary pass/fail signal in classical MDS-based DAS.

3.4 Interpretation of Recovery Thresholds

Ambiguity often arises in data availability discussions regarding the number of symbols required for recovery, as this depends on the specific adversarial model. We therefore separate three distinct notions.

(1) *Dimension k (existential information sets)*. For any (n, k, d) code, the message has k degrees of freedom. In particular, there exist *information sets* of size k whose values uniquely determine the message. For systematic encodings, there is an explicit set of k coordinates containing the message symbols. However, for random sampling, hitting a specific information set is not guaranteed, and for adversarial erasures, a set of size k may fail to be sufficient if it is not an information set.

(2) *Distance-based erasure recovery $(n - d + 1)$ (worst-case guarantee)*. For any code of minimum distance d , *any* set of at least $n - d + 1$ known symbols uniquely determines the entire codeword under erasures. This is a worst-case statement: it holds for *every* erasure pattern of size at most $d - 1$.

(3) *Polynomial interpolation $(D + 1)$ (degree-based sufficient condition)*. In our evaluation-code setting, codewords arise as evaluations of a global polynomial $f(X)$ of degree at most D (Section 2.3). Hence, any set of $D + 1$ *distinct, correct* evaluations determines f uniquely, and therefore determines the entire evaluation vector on A . This is a sufficient condition tied to the chosen degree bound, and should not be conflated with the existential minimum k .

Implication for LR-DAS. LR-DAS certifies availability progressively via local groups: each locally certified group yields $r + \alpha$ determined symbols. Full reconstruction can be triggered once the number of determined symbols crosses a global recovery threshold of interest, e.g., $n - d + 1$ for worst-case erasures or $D + 1$ for degree-based interpolation in our polynomial setting. Later sections analyze how sampling budgets translate into progress in $A(S)$ and into these global thresholds.

3.5 Sampling policies and distinct hits per group

Let Q be the sampled multiset of size s and define, for each group j ,

$$S_j := |\text{supp}(Q \cap A_j)|, \quad (11)$$

the number of *distinct* sampled points in group A_j . A necessary condition for certifying A_j is $S_j \geq r$. Group-aware sampling can be used to control the distribution of S_j across groups (by enforcing per-group budgets), while uniform sampling may lead to imbalanced coverage for small budgets. Our graded-soundness bounds later separate (i) the probability that $S_j \geq r$ in withheld groups and (ii) the probability of any cryptographic acceptance error of the underlying commitment interface.

4 Locality-Aware Erasure-Code Commitment

This section defines the commitment primitive needed by LR-DAS. Intuitively, the verifier wants to *certify* a local group A_j after learning only a few openings in that group. To make such certification meaningful, openings must be bound to a *single* globally committed LRC codeword, and (crucially) a locally certified group must be extendable to its full $(r + \alpha)$ symbols in a way that is consistent with the same global codeword. We formalize this as a locality-aware variant of erasure-code commitments, building on the framework of [7].

4.1 Challenges in Locality-Aware Commitments

Local recovery groups enable graded availability: once r correct symbols from a group A_j are known, the entire group can be reconstructed by interpolating a unique degree- $< r$ polynomial on A_j . However, *local* structure alone does not prevent *splicing*: an adversary could attempt to provide locally consistent groups that do not arise as restrictions of any *single* global codeword in the designated LRC code C . Such splicing invalidates graded certification, because the verifier may reconstruct a group using a local witness that is inconsistent with the globally committed object. Therefore, LR-DAS requires a commitment notion that simultaneously provides: (i) *global binding* to one LRC codeword and (ii) *local-global consistency* for every accepted local certificate.

4.2 Syntax

Fix public parameters $(\mathbb{F}_q, A = \bigsqcup_{j=0}^{\ell-1} A_j, r, \alpha)$ and a designated LRC code $C \subseteq \mathbb{F}_q^n$ with disjoint recovery groups (Section 2.2). A locality-aware erasure-code commitment provides openings for subsets within a group, and may additionally support out-of-domain openings for auxiliary checks (used later to enforce local-global consistency succinctly).

Definition 1 (Locality-aware erasure-code commitment). A locality-aware erasure-code commitment for $(C, A, \{A_j\}, r, \alpha)$ consists of PPT algorithms:

$$\begin{aligned} \text{Setup}(1^\lambda) &\rightarrow \text{ck}, \\ \text{Com}(\text{ck}, m) &\rightarrow (\text{com}, \text{St}), \\ \text{OpenLocal}(\text{ck}, \text{St}, j, U) &\rightarrow (\{(a, v_a)\}_{a \in U}, \pi), \\ \text{VerifyLocal}(\text{ck}, \text{com}, j, U, \{(a, v_a)\}_{a \in U}, \pi) &\in \{0, 1\}, \end{aligned}$$

where $j \in \{0, \dots, \ell - 1\}$ and $U \subseteq A_j$.

Optionally, the scheme may also expose point openings of a globally committed auxiliary object (e.g., a polynomial) at any $x \in \mathbb{F}_q$ (including $x \notin A$):

$$\text{OpenPoint}(\text{ck}, \text{St}, x) \rightarrow (y, \pi_x), \quad \text{VerifyPoint}(\text{ck}, \text{com}, x, y, \pi_x) \in \{0, 1\}.$$

For $x \in A$, the intended meaning is that y equals the committed codeword symbol at position x . For $x \notin A$, the value y is auxiliary and will be used to enforce local-global consistency efficiently.

Correctness. For any honestly generated $(\text{com}, \text{St}) \leftarrow \text{Com}(\text{ck}, m)$, any j , and any $U \subseteq A_j$, the honest opening produced by `OpenLocal` verifies under `VerifyLocal`. Similarly, if the optional point-opening interface is provided, honest point openings verify under `VerifyPoint`.

4.3 Security definitions

We next formalize the properties needed by LR-DAS. The definitions are phrased so that (i) the phrase “single global codeword” is unambiguous and (ii) trivial per-group constructions (e.g., committing independently to each group) are ruled out. Our main notion is a reconstruction-binding style guarantee, as advocated for DAS in [7], augmented with an explicit local-global consistency requirement.

Position binding (no equivocation on a coordinate). Position binding is the standard DAS requirement: after a commitment is fixed, no adversary should be able to open the same *domain point* $a \in A$ to two different values.

Definition 2 (Position binding). A *locality-aware erasure-code commitment* is ε_{pos} -position binding if for every PPT adversary $\mathcal{A}$:

$$\Pr \left[\begin{array}{l} \text{VerifyPoint}(\text{ck}, \text{com}, a, y, \pi) = 1 \\ \wedge \text{VerifyPoint}(\text{ck}, \text{com}, a, y', \pi') = 1 \end{array} \middle| \begin{array}{l} \text{ck} \leftarrow \text{Setup}(1^\lambda), \\ (\text{com}, i, (y, \pi), (y', \pi')) \leftarrow \mathcal{A}(\text{ck}) \end{array} \right] \leq \varepsilon_{\text{pos}}$$

where $y \neq y'$. If the scheme does not expose `OpenPoint/VerifyPoint`, the same definition applies with `VerifyLocal` using singleton sets $U = \{a\} \subseteq A_j$ (with the unique j such that $a \in A_j$).

Local binding (unique local polynomial per group). Local binding ensures that a commitment fixes a unique local polynomial for each group. Specifically, for a fixed commitment com and a group index j , it should be computationally infeasible for an adversary to produce two sets of valid local openings that cannot be interpolated by a *single* polynomial $\delta_j(X)$ of degree less than r .

One can interpret this property as a notion of local level *code-binding*: it ensures that the openings within a group A_j are consistent with a valid codeword of the underlying local code, effectively applying the code-binding requirement at the granularity of a local group.

Definition 3 (Local binding). A *locality-aware erasure-code commitment* is ε_{loc} -locally binding if for every PPT adversary $\mathcal{A}$:

$$\Pr \left[\begin{array}{l} \text{VerifyLocal}(\text{ck}, \text{com}, j, U_1, S_1, \pi_1) = 1 \\ \wedge \text{VerifyLocal}(\text{ck}, \text{com}, j, U_2, S_2, \pi_2) = 1 \\ \wedge \nexists \delta(X) \in \mathbb{F}_q[X] \text{ s.t. } \deg(\delta) < r \\ \text{and } \forall (a, v) \in S_1 \cup S_2, \delta(a) = v \end{array} \middle| \begin{array}{l} \text{ck} \leftarrow \text{Setup}(1^\lambda), \\ (\text{com}, j, \mathcal{O}_1, \mathcal{O}_2) \leftarrow \mathcal{A}(\text{ck}) \end{array} \right] \leq \varepsilon_{\text{loc}}$$

where for each $k \in \{1, 2\}$, $\mathcal{O}_k = (U_k, S_k, \pi_k)$ is a local opening on the group A_j (i.e., $U_k \subseteq A_j$) with the openings $(S_k = \{(a, v_a)\}_{a \in U_k}, \pi_k)$.

Local-global consistency (no cross-group splicing). Local binding alone implies only that openings within a group are self-consistent, but it does not prevent *splicing* attacks where an adversary opens values consistent locally but inconsistent with the globally committed codeword. To rule this out, we require that any accepted local opening for a coordinate $a \in A_j$ must yield a value identical to the one obtained from the global commitment at the same coordinate.

Definition 4 (Local-global consistency). A locality-aware erasure-code commitment is ε_{lg} -locally-globally consistent if for every PPT adversary $\mathcal{A}$:

$$\Pr \left[\begin{array}{l} \text{VerifyLocal}(\text{ck}, \text{com}, j, \{a\}, S_L, \pi_L) = 1 \\ \wedge \text{VerifyPoint}(\text{ck}, \text{com}, a, y, \pi_a) = 1 \\ \wedge v \neq y \end{array} \middle| \begin{array}{l} \text{ck} \leftarrow \text{Setup}(1^\lambda), \\ (\text{com}, j, \mathcal{O}_L, \mathcal{O}_G) \leftarrow \mathcal{A}(\text{ck}) \end{array} \right] \leq \varepsilon_{\text{lg}}$$

where $\mathcal{O}_L = (S_L, \pi_L)$ is a local opening for a point $a \in A_j$ with value set $S_L = \{(a, v)\}$, and $\mathcal{O}_G = (a, y, \pi_a)$ is a global opening at the same a .

Reconstruction binding to a single LRC codeword. The previous notions address consistency of openings, but LR-DAS additionally needs that commitments bind to the *designated* LRC code C (e.g., the TB code), not merely to an arbitrary vector in $\mathbb{F}_q^n$. Because DAS relies on reconstructing the original data from a subset of samples, it is a core requirement that the commitment is strictly binding to a single global codeword. We adopt an extractor-based formulation (reconstruction binding) in the spirit of [7].

Definition 5 (Reconstruction binding to C). A locality-aware erasure-code commitment is ε_{rb} -reconstruction binding for the code C if there exists a PPT extractor $\mathcal{E}$ such that for every PPT adversary $\mathcal{A}$, the following holds:

$$\Pr \left[\begin{array}{l} \text{VerifyPoint}(\text{ck}, \text{com}, a, v, \pi) = 1 \\ \wedge (c^* \notin C \vee v \neq c^*(a)) \end{array} \middle| \begin{array}{l} \text{ck} \leftarrow \text{Setup}(1^\lambda), \\ (\text{com}, a, v, \pi) \leftarrow \mathcal{A}(\text{ck}), \\ c^* \leftarrow \mathcal{E}^{\mathcal{A}}(\text{ck}, \text{com}) \end{array} \right] \leq \varepsilon_{\text{rb}}$$

where $\mathcal{E}$ has rewind access to $\mathcal{A}$.

Equivalently, this guarantees that except with probability ε_{rb} over the randomness of $\mathcal{E}$, $\mathcal{A}$, and Setup , any accepting opening produced by $\mathcal{A}$ at any domain point $a \in A$ is strictly consistent with a single codeword c^* in the designated LRC code C .

Implication for LR-DAS certification. Assuming reconstruction binding to C and local binding (Definitions 3 and 5), once the verifier has obtained r accepted distinct openings in a group A_j , the unique degree- $< r$ polynomial interpolating these points equals the restriction of the extracted codeword c^* to A_j . Hence reconstructing the remaining α symbols in A_j is well-defined and consistent with the same global codeword. Local-global consistency (Definition 4) makes this compatibility explicit at the interface level and rules out cross-interface splicing. Later sections show how to realize these properties efficiently using a single out-of-domain algebraic link.

5 Special constraint for LRC code commitment

This section presents the core algebraic mechanism that enforces local–global consistency with a *single* out-of-domain check per certified local group. The purpose of the construction is to rule out cross-group splicing: a prover should not be able to provide locally consistent witnesses for multiple groups that do not arise as restrictions of one globally committed codeword. Our definitions in Section 4.3 make this requirement explicit; here we show how to realize it.

5.1 Algebraic form of local–global agreement

Fix a group index $j \in \{0, \dots, \ell - 1\}$. Let $A_j \subseteq A$ be the local group of size $|A_j| = r + \alpha$. Recall the vanishing polynomial notation $Z_S(X) = \prod_{a \in S} (X - a)$ from (1).

Global and local polynomials. We consider:

- a *global* polynomial $f(X) \in \mathbb{F}_q[X]$ of degree at most D whose evaluations on A define the committed codeword, and
- a *local* polynomial $\delta_j(X) \in \mathbb{F}_q[X]$ of degree $< r$ intended to equal the restriction of f on A_j .

Define the group vanishing polynomial

$$h_j(X) := Z_{A_j}(X) = \prod_{a \in A_j} (X - a), \quad (12)$$

so $\deg(h_j) = r + \alpha$ and $h_j(a) = 0$ for all $a \in A_j$.

Divisibility characterization. The condition “ δ_j equals f on A_j ” is equivalent to a divisibility statement.

Lemma 1 (Restriction $\Leftrightarrow$ divisibility). *For any polynomials $f, \delta_j \in \mathbb{F}_q[X]$, the following are equivalent:*

1. $f(a) = \delta_j(a)$ for all $a \in A_j$ (i.e., $f|_{A_j} = \delta_j|_{A_j}$ as functions);
2. $h_j(X)$ divides $f(X) - \delta_j(X)$ in $\mathbb{F}_q[X]$, i.e., there exists a polynomial $q_j(X)$ such that

$$f(X) - \delta_j(X) = h_j(X) \cdot q_j(X). \quad (13)$$

Proof. If (13) holds, then for every $a \in A_j$ we have $h_j(a) = 0$, hence $f(a) - \delta_j(a) = 0$. Conversely, if $f(a) = \delta_j(a)$ for all $a \in A_j$, then $f - \delta_j$ has all points in A_j as roots, hence it is divisible by $\prod_{a \in A_j} (X - a) = h_j(X)$.

Thus, enforcing (13) is sufficient to guarantee local–global consistency for group A_j . The challenge is to enforce this identity efficiently without checking all $r + \alpha$ points in A_j .

5.2 A one-point out-of-domain linking check

We now show that (13) can be enforced by checking it at a *single* random out-of-domain point $\rho \in \mathbb{F}_q \setminus A$. This is a standard “out-of-domain” identity test in the spirit of DEEP-style checks used with FRI/IOPP systems [3].

Degree bounds. Let D_f be a known upper bound on $\deg(f)$ and let D_q be a known upper bound on $\deg(q_j)$. In our applications, we will set $D_f = D$ and enforce $D_q \leq D - (r + \alpha)$ so that $\deg(h_j \cdot q_j) \leq D$. Let

$$D_{\text{link}} := \max\{D_f, r - 1, (r + \alpha) + D_q\}. \quad (14)$$

Commit-before-challenge requirement. A crucial point for soundness is that the prover must not be able to choose q_j *after* learning ρ . In an interactive protocol, this is ensured by sending commitments to f , δ_j , and q_j before ρ is chosen. In a Fiat–Shamir (FS) transform, this is ensured by deriving ρ as a hash of the transcript that includes the commitments to f , δ_j , and q_j . Otherwise, a cheating prover could always set $q_j(\rho) = (f(\rho) - \delta_j(\rho))/h_j(\rho)$ and pass the check without implying (13).

Lemma 2 (One-link soundness (one-point identity test)). *Fix polynomials $f, \delta_j, q_j \in \mathbb{F}_q[X]$ with $\deg(f) \leq D_f$, $\deg(\delta_j) \leq r - 1$, and $\deg(q_j) \leq D_q$. Define*

$$L_j(X) := f(X) - \delta_j(X) - h_j(X) q_j(X).$$

Assume commitments to f , δ_j , and q_j are fixed before sampling a challenge point $\rho \leftarrow \mathbb{F}_q \setminus A$ uniformly at random. Then:

1. *If $L_j(X) \equiv 0$ (as a polynomial), then the verifier accepts the link check at every ρ .*
2. *If $L_j(X) \not\equiv 0$, then*

$$\Pr_{\rho \leftarrow \mathbb{F}_q \setminus A} [L_j(\rho) = 0] \leq \frac{D_{\text{link}}}{q - |A|}. \quad (15)$$

Consequently, except with probability at most $D_{\text{link}}/(q - |A|)$, acceptance of the one-point check implies the polynomial identity (13), and hence $f|_{A_j} = \delta_j|_{A_j}$ by Lemma 1.

Proof. If $L_j \equiv 0$ then $L_j(\rho) = 0$ for all ρ and the check always passes. Otherwise, L_j is a nonzero polynomial of degree at most D_{link} by (14), and it has at most D_{link} roots in $\mathbb{F}_q$. Sampling ρ uniformly from $\mathbb{F}_q \setminus A$ yields (15). The final implication follows from the definition of L_j and Lemma 1.

Interpretation. The one-link check enforces that every locally accepted witness δ_j is consistent with the *same* global polynomial f . This directly prevents cross-group splicing at the level required by the local-global consistency definition (Definition 4).

Link-Prove(ck, f, δ_j) for group A_j :

1. Compute $h_j(X) = Z_{A_j}(X)$.
2. Compute a polynomial $q_j(X)$ such that $f(X) - \delta_j(X) = h_j(X) \cdot q_j(X)$ (if it exists).
3. Commit to f , δ_j , and q_j (or include their commitments in the global commitment transcript).
4. Derive an out-of-domain challenge $\rho \in \mathbb{F}_q \setminus A$ from the transcript (interactive: verifier samples; FS: ρ is a hash of commitments, including the commitment to q_j).
5. Open $f(\rho)$, $\delta_j(\rho)$, and $q_j(\rho)$ and provide the corresponding opening proofs.

Link-Verify for group A_j :

1. Recompute $h_j(\rho) = Z_{A_j}(\rho)$ and verify the three point openings at ρ .
2. Accept iff $f(\rho) - \delta_j(\rho) = h_j(\rho) \cdot q_j(\rho)$.

Fig. 2. The one-link local-global binding check. Soundness follows from Lemma 2.

Corollary 1 (Link $\Rightarrow$ interface-level local-global consistency). *Assume that, for a fixed commitment com, the verifier accepts the one-link check for group A_j with respect to some committed polynomials (f, δ_j, q_j) . Then for every $a \in A_j$, $f(a) = \delta_j(a)$. In particular, any accepted local opening value at a point $a \in A_j$ equals the value of the globally committed codeword at a .*

Proof. By Lemma 2, acceptance implies (13) except with probability $D_{\text{link}}/(q - |A|)$, hence $f(a) = \delta_j(a)$ for all $a \in A_j$ by Lemma 1.

5.3 A generic linking subprotocol

We now state the linking check as a small subprotocol that can be instantiated by any polynomial commitment scheme supporting point openings (e.g., KZG [10] or transparent PCS stacks based on FRI [3]). This subprotocol is used as a component inside the locality-aware commitment of Section 4 and inside LR-DAS certification (Section 3.3).

Remark: TB Membership versus Restriction. The link enforces that a local witness δ_j equals the restriction of the committed global polynomial f on A_j . This does *not* by itself ensure that f corresponds to the intended structured LRC code C (e.g., TB membership); that global membership/binding is provided by the overall commitment scheme (Definition 5) and, in our instantiations, by either construction-time encoding or an outer proximity check. The link acts as the minimal binding mechanism that prevents *splicing across groups* once a single global object is fixed.

5.4 Using the link for LR-DAS group certification

We summarize how the link integrates into LR-DAS certification of a group A_j .

Inputs to certification. For a fixed group A_j , the verifier collects accepted local openings on a set $U \subseteq A_j$ with $|U| \geq r$. The locality-aware commitment must ensure: (i) the opened values are consistent with some degree- $< r$ polynomial δ_j (local membership), and (ii) the one-link check binds δ_j to the global polynomial f (Figure 2).

Outcome. Once both conditions hold, the verifier interpolates δ_j from any r points in U and reconstructs the entire group by evaluating δ_j on all of A_j . By Corollary 1, these reconstructed values equal the restriction of the global codeword bound by com .

5.5 Batching within a group and soundness amplification

Batching openings inside a group (optional but useful). If the verifier requests many points in a group A_j , it is convenient to open them with a single batched proof. A standard method uses the vanishing polynomial $Z_U(X)$ for a set U of query points and proves the identity

$$f(X) = R_U(X) + Z_U(X) \cdot W_U(X),$$

where $R_U(X) = f(X) \bmod Z_U(X)$ is the remainder (degree $< |U|$) that encodes the opened evaluations. This technique yields efficient multipoint openings for polynomial commitments [4] and applies directly to opening δ_j on $U \subseteq A_j$.

Amplifying the one-link error and field-size considerations. Lemma 2 shows that the one-link check fails with probability at most $D_{\text{link}}/(q - |A|)$. If this is not negligible for a given parameter set (e.g., due to a modest field size), the error can be amplified by checking the link at t independent out-of-domain points $\rho_1, \dots, \rho_t$ derived from independent domain-separated challenges: the soundness error becomes at most $(D_{\text{link}}/(q - |A|))^t$. Alternatively, one can work over an extension field of larger size. These choices do not affect the LR-DAS syntax; they only trade off proof size and verification work against link soundness.

6 Graded Soundness

This section formalizes and proves the graded-soundness guarantees of LR-DAS. Our central result establishes that *false graded certification* (“overshoot”) decomposes exactly into: (i) a *sampling term* capturing how likely the verifier is to probe an unavailable group enough times, and (ii) a *cryptographic term* capturing the probability that the commitment verification accepts a false local certificate. This style of decomposition is standard in the erasure-code commitment view of DAS [7], but LR-DAS requires a locality-aware formulation because certification is performed at the granularity of recovery groups.

6.1 Missing patterns and which groups are in principle certifiable

LR-DAS certifies groups using only local recovery inside each A_j : a group is certified only after the verifier obtains and verifies r *distinct* openings in that group (Section 3.3). Therefore, the natural “ground truth” for graded availability is whether a group has at least r symbols that are actually retrievable (so that local interpolation is possible).

Missing set and induced unavailable groups. Let $W \subseteq A$ be a set of *missing* (unavailable) coordinates. We model W as the set of domain points for which no honest storage node can answer with a valid opening. For each group A_j , define the number of available coordinates

$$\text{avail}(j; W) := |A_j \setminus W|.$$

We say that group A_j is *locally recoverable under W* if $\text{avail}(j; W) \geq r$ (equivalently, $|A_j \cap W| \leq \alpha$ since $|A_j| = r + \alpha$). Define the sets of recoverable and unrecoverable groups:

$$\begin{aligned} \text{Good}(W) &:= \{j : \text{avail}(j; W) \geq r\} \\ \text{Bad}(W) &:= [\ell] \setminus \text{Good}(W) \\ g(W) &:= |\text{Bad}(W)| \end{aligned} \tag{16}$$

Groups in $\text{Bad}(W)$ have fewer than r retrievable symbols and hence cannot be reconstructed locally.

Attainable graded availability under W . Since LR-DAS certifies groups via r distinct verified openings in that same group, no algorithm can *soundly* certify any $A_j \in \text{Bad}(W)$ without accepting a forged opening for at least one missing point. Therefore, the maximal *sound* certified fraction under a missing pattern W is at most

$$\phi_{\max}(W) := \frac{|\text{Good}(W)|}{\ell} = 1 - \frac{g(W)}{\ell}.$$

Our graded-soundness statements will bound the probability that the verifier outputs a fraction exceeding $\phi_{\max}(W)$.

Special case: missing entire groups. If W is a union of whole groups, i.e., $W = \bigcup_{j \in M} A_j$ for some $M \subseteq [\ell]$, then $g(W) = |M|$ and $\phi_{\max}(W) = (\ell - |M|)/\ell$. This yields the canonical “missing-groups” adversarial model used throughout the paper.

6.2 Security definitions: overshoot and graded soundness

Recall that LR-DAS outputs a certified fraction $\phi(S) = A(S)/\ell$ as defined in (10).

Definition 6 (Overshoot under a missing pattern). Fix a missing set $W \subseteq A$ and let $\phi_{\max}(W) = |\text{Good}(W)|/\ell$. We say that an LR-DAS execution overshoots under W if the verifier outputs

$$\phi(S) > \phi_{\max}(W).$$

Equivalently, overshoot occurs iff the verifier certifies at least one $A_j \in \text{Bad}(W)$.

Definition 7 (Graded soundness: missing-groups model). For an integer $g \in \{1, \dots, \ell\}$, LR-DAS is $(\varepsilon_{\text{grp}}, g)$ -graded sound (missing-groups) if for every PPT adversary that induces a missing set $W = \bigcup_{j \in M} A_j$ with $|M| \geq g$, the probability that the verifier overshoots (Definition 6) is at most ε_{grp} .

Definition 8 (Graded soundness: missing-fraction model (via induced bad groups)). For $X \in (0, 1)$, LR-DAS is $(\varepsilon_{\text{frac}}, X)$ -graded sound (missing-fraction) if for every PPT adversary that induces a missing set $W \subseteq A$ with $|W| \geq Xn$, the probability of overshoot under W is at most $\varepsilon_{\text{frac}}$.

Remark (why the definition is group-based). Because LR-DAS local recovery tolerates up to α missing symbols per group, a coordinate-level missing fraction by itself does not uniquely determine how many groups are unrecoverable: if missing symbols are spread so that each group loses at most α symbols, then all groups remain locally recoverable. Definition 8 therefore measures overshoot relative to $\phi_{\max}(W)$ induced by the actual missing pattern, rather than using an invalid bound of the form “ $1 - X$ ”.

6.3 From missing coordinates to missing groups

The following simple counting lemma connects a coordinate-level missing budget to a guaranteed number of unrecoverable groups.

Lemma 3 (Lower bound on unrecoverable groups from $|W|$). Let $W \subseteq A$ be any missing set and let $g(W)$ be defined as in (16). Then

$$|W| \leq \alpha\ell + r \cdot g(W), \text{ and hence } g(W) \geq \max\left\{0, \left\lceil \frac{|W| - \alpha\ell}{r} \right\rceil\right\}. \quad (17)$$

Proof. For any group index $j \in \text{Good}(W)$ we have $|A_j \cap W| \leq \alpha$. For any $j \in \text{Bad}(W)$ we have the trivial upper bound $|A_j \cap W| \leq |A_j| = r + \alpha$. Summing over groups,

$$\begin{aligned} |W| &= \sum_{j \in \text{Good}(W)} |A_j \cap W| + \sum_{j \in \text{Bad}(W)} |A_j \cap W| \\ &\leq \alpha \cdot |\text{Good}(W)| + (r + \alpha) \cdot |\text{Bad}(W)|. \end{aligned}$$

Since $|\text{Good}(W)| = \ell - g(W)$, this equals $\alpha(\ell - g(W)) + (r + \alpha)g(W) = \alpha\ell + rg(W)$, proving the first inequality. Rearranging yields the lower bound on $g(W)$.

Corollary (a valid “missing-fraction $\Rightarrow$ at least g bad groups” statement). If $|W| \geq Xn$ and $n = (r + \alpha)\ell$, then Lemma 3 implies

$$g(W) \geq \max \left\{ 0, \left\lceil \frac{X(r + \alpha)\ell - \alpha\ell}{r} \right\rceil \right\}.$$

In particular, if $X \leq \alpha/(r + \alpha)$ this lower bound is 0, reflecting that up to an $\alpha/(r + \alpha)$ missing fraction can still leave all groups locally recoverable (when losses are diffuse).

6.4 Overshoot bounds: sampling term times cryptographic term

We now state the main bound. Fix a sampling policy (Section 3.2) and let Q be the sampled multiset of size s . Recall $S_j = |\text{supp}(Q \cap A_j)|$ from (11), the number of *distinct* sampled points in group A_j .

Cryptographic failure parameter. Let $\varepsilon_\star$ be an upper bound on the probability that the locality-aware commitment falsely certifies an unrecoverable group once the verifier has sampled enough distinct points. Concretely, for any group $A_j \in \text{Bad}(W)$ and any set $U \subseteq A_j$ with $|U| \geq r$, $\varepsilon_\star$ bounds the probability that the verifier accepts the local opening and the local-global consistency checks for that group despite the fact that U necessarily contains missing points. In our instantiations, $\varepsilon_\star$ is derived from: (i) the soundness of the local degree- $< r$ membership checks and (ii) the soundness of the one-link out-of-domain check (Section 5); see Section 7.

Theorem 1 (Graded soundness via overshoot bound). *Fix a missing set $W \subseteq A$ and let $\text{Bad}(W)$ be defined by (16). Assume non-adaptive withholding: W is fixed before the verifier’s sampling randomness is determined. For each group index j , define the sampling-success probability*

$$p_j := \Pr[S_j \geq r],$$

where the probability is over the verifier’s sampling policy (and its randomness).

Assume that, for every $j \in \text{Bad}(W)$, conditioned on $S_j \geq r$, the probability that the verifier accepts group A_j as locally certified is at most $\varepsilon_\star$. Let $\varepsilon_{\text{glob}}$ be an upper bound on any remaining global failure event (e.g., failure of global binding/position binding of the overall commitment). Then the overshoot probability satisfies

$$\Pr[\phi(S) > \phi_{\max}(W)] \leq \sum_{j \in \text{Bad}(W)} p_j \cdot \varepsilon_\star + \varepsilon_{\text{glob}}. \quad (18)$$

In particular, in the missing-groups model with $W = \bigcup_{j \in M} A_j$, we have $\text{Bad}(W) = M$ and

$$\Pr \left[\phi(S) > \frac{\ell - |M|}{\ell} \right] \leq \sum_{j \in M} p_j \cdot \varepsilon_\star + \varepsilon_{\text{glob}}.$$

Proof (Proof sketch). If $\phi(S) > \phi_{\max}(W)$, then at least one group in $\text{Bad}(W)$ was accepted as locally certified. For a fixed $j \in \text{Bad}(W)$, acceptance implies both: (i) $S_j \geq r$ (the verifier sampled at least r distinct points in that group), and (ii) the commitment checks for that group accepted on a set that necessarily includes missing coordinates. By the non-adaptive withholding assumption, the sampling event is independent of the commitment’s soundness error, so $\Pr[A_j \text{ accepted}] \leq p_j \cdot \varepsilon_\star$. A union bound over $j \in \text{Bad}(W)$ yields the first term in (18), and adding the global failure probability $\varepsilon_{\text{glob}}$ gives the claim.

6.5 Computing the sampling term p_j (group-aware sampling)

Theorem 1 separates sampling from cryptography. Here we give an explicit expression for p_j under a common group-aware sampling policy.

Per-group sampling with replacement. Suppose the verifier assigns a budget s_j to group A_j and samples s_j times independently and uniformly from A_j (with replacement). Let $m := |A_j| = r + \alpha$. Then the distribution of S_j (the number of distinct sampled points in A_j) is the standard occupancy distribution: for $t \in \{0, 1, \dots, m\}$,

$$\Pr[S_j = t] = \binom{m}{t} t! \left\{ \begin{matrix} s_j \\ t \end{matrix} \right\} m^{-s_j},$$

where $\left\{ \begin{matrix} s \\ t \end{matrix} \right\}$ denotes the Stirling number of the second kind (the number of partitions of s labeled draws into t nonempty bins). Therefore,

$$p_j = \Pr[S_j \geq r] = 1 - \sum_{t=0}^{r-1} \binom{r+\alpha}{t} t! \left\{ \begin{matrix} s_j \\ t \end{matrix} \right\} (r+\alpha)^{-s_j}. \quad (19)$$

Uniform sampling over A can be handled similarly (via coupling or direct occupancy bounds), but we omit these routine details.

Connecting to link soundness and field size. In our one-link construction (Section 5), the local–global linking mechanism is a one-point out-of-domain identity test. Its algebraic soundness term is at most $D_{\text{link}}/(q - |A|)$ as in Lemma 2 and can be amplified by checking multiple out-of-domain points (Section 5.5). These terms are absorbed into $\varepsilon_\star$ and made negligible for concrete parameter choices (Section 7).

7 Instantiations: KZG and FRI

We instantiate the locality-aware erasure-code commitment interface of Section 4 and the one-link local–global binding mechanism of Section 5 in two complementary ways: (i) a succinct pairing-based instantiation using KZG polynomial commitments [10], and (ii) a transparent instantiation based on Merkle commitments and FRI/IOPP low-degree tests [3,8]. Both instantiations support multi-point openings via standard vanishing-polynomial batching [4].

Table 2. Trade-offs between instantiations. “Link” refers to the one-point out-of-domain identity test from Section 5.

	KZG-based	FRI-based
Setup	Structured SRS	Transparent Setup
Proof size	$O(1)$	$\text{polylog}(n)$
Verifier work	$O(1)$	$\text{polylog}(n)$
Main assumptions	Cryptographic pairing, KZG binding	Hash collision resistance, FRI soundness
Soundness components	KZG binding + link	Inner/outer LDTs + link

7.1 KZG-based locality-aware commitment (succinct, two-layer)

Coding layer (systematic TB setting). This instantiation targets the *systematic* TB regime $k = \ell r$ (so each of the r coefficient functions has full dimension ℓ over the ℓ local groups). Let $g(X)$ be a good polynomial that is constant on each group A_j with distinct values $\gamma_j := g(a)$ for any $a \in A_j$ (Section 2.3). Let $L_0, \dots, L_{\ell-1} \in \mathbb{F}_q[T]$ be the Lagrange basis on the points $\{\gamma_0, \dots, \gamma_{\ell-1}\}$, i.e., $L_j(\gamma_i) = \mathbf{1}\{i = j\}$. Define selectors

$$s_j(X) := L_j(g(X)). \quad (20)$$

Then $s_j(a) = 1$ for $a \in A_j$ and $s_j(a) = 0$ for $a \in A_i$ with $i \neq j$. Given local polynomials $\delta_0, \dots, \delta_{\ell-1}$ with $\deg(\delta_j) < r$, define the assembled global polynomial

$$f(X) := \sum_{j=0}^{\ell-1} s_j(X) \delta_j(X), \quad (21)$$

which satisfies $f|_{A_j} = \delta_j|_{A_j}$ for all j . In this systematic setting, $(f(a))_{a \in A}$ is a TB codeword (Section 2.3), and local recovery inside a certified group is simply interpolation of δ_j .

Cryptographic layer. We use KZG commitments [10] for the global polynomial f and each local polynomial δ_j . To minimize prover overhead, local openings use δ_j (degree $< r$) rather than the global f (degree $\leq D$), and a one-point link enforces that the opened local view equals the restriction of f .

Security accounting (connection to Section 6). Let ε_{KZG} bound the probability of breaking KZG evaluation binding for either C_{glob} or any $C_{\text{loc}}^{(j)}$ (under the usual pairing-based assumptions for KZG [10]). Let $\varepsilon_{\text{link}}$ be the algebraic soundness error of the one-link check from Lemma 2 (e.g., $\varepsilon_{\text{link}} \leq D_{\text{link}}/(q - |A|)$), or amplified by repeating the link at multiple points). Then a conservative choice consistent with Theorem 1 is

$$\varepsilon_{\star}^{\text{KZG}} \leq \varepsilon_{\text{KZG}} + \varepsilon_{\text{link}}, \quad \varepsilon_{\text{glob}}^{\text{KZG}} \leq \varepsilon_{\text{KZG}}. \quad (22)$$

Intuitively, local certification of an *unrecoverable* group requires either a false KZG acceptance or a false one-link acceptance.

Summary of KZG-LAECC (Details in Appendix Fig. 5)

- **Setup:** Generates dual SRS ($\text{SRS}_{\text{glob}}, \text{SRS}_{\text{loc}}$).
- **Com:** Commits to global polynomial f , local δ_j , and quotient q_j .
- **OpenLocal:**
 - Opens δ_j at subset of local group (i.e., $U \subseteq A_j$) yields openings V_U .
 - Opens f, δ_j, q_j at a commitment-derived challenge ρ .
- **VerifyLocal:**
 - Verifies all openings V_U .
 - Verify *One-Link check* constraint.
 - Verify *One-Link check* points.

Fig. 3. Workflow of the KZG-based LAECC.**7.2 FRI-based locality-aware commitment (transparent IOPP stack)**

High-level structure. We use a transparent commitment stack in which: (i) a Merkle root commits to an evaluation table, (ii) FRI-style low-degree tests (LDTs) provide proximity to low-degree polynomials, and (iii) a single out-of-domain one-link check ties the local and global views. This is consistent with the IOPP/FRI viewpoint of [3] and the DAS-oriented PCS approach in [8].

Committed object. Let $f(X)$ be the TB encoding polynomial (Section 2.3) and define the table $D \in \mathbb{F}_q^{\ell \times (r+\alpha)}$ by rows $D_{j,*} = (f(a))_{a \in A_j}$. We commit to D with a Merkle tree whose leaves are row hashes (any standard collision-resistant hash suffices). The Merkle root serves as the public commitment to the polynomial within the FRI proof.

Inner vs. outer tests. The proof has three logically separate parts:

1. **Inner (local) LDT:** for each queried group j , the prover convinces the verifier that the committed row $D_{j,*}$ is the evaluation of some degree- $< r$ polynomial δ_j on the points A_j .
2. **Outer (global) LDT:** the prover convinces the verifier that the entire table D is close to a *single* TB codeword, i.e., to evaluations of one global low-degree object with TB structure (global binding to the designated code).
3. **One-link check:** the prover provides out-of-domain evaluations at a fresh $\rho \notin A$ for both the global and local views and proves the identity $f(\rho) - \delta_j(\rho) = h_j(\rho)q_j(\rho)$, as in Section 5.

The outer test can be verified once per commitment, while the inner+link checks are performed for certified groups only (and can be batched across many groups using standard FRI batching techniques).

Security accounting. Let $\varepsilon_{\text{Merkle}}$ bound the hash collision probability (position binding for committed rows). Let ε_{out} be the soundness error of the outer proximity proof Π_{out} (binding to a single global TB codeword), and let ε_{in} be the

Summary of FRI-LAECC (Details in Appendix Fig. 6)

- **Setup:** Fix hash functions $(\mathcal{H}_{\text{Merkle}}, \mathcal{H}_{\text{FS}})$.
- **Com:**
 - Generate FRI proofs of global polynomial f , local δ_j , and quotient q_j .
- **OpenLocal:**
 - Opens δ_j at subset of local group (i.e., $U \subseteq A_j$) yields openings V_U .
 - Calculate corresponding Merkle paths to $R_f, R_{\delta_j}, R_{q_j}$.
 - Opens f, δ_j, q_j at a commitment-derived challenge ρ .
- **VerifyLocal:**
 - Verifies all openings V_U via Merkle tree.
 - Verify *One-Link check* constraint.
 - Verify all FRI proofs.

Fig. 4. Workflow of the FRI-based LAECC.

soundness error of each inner row test (binding the row to a degree- $< r$ polynomial). Let $\varepsilon_{\text{link}}$ be the one-link algebraic error (Lemma 2), with optional amplification as in Section 5.5. A conservative choice matching Theorem 1 is

$$\varepsilon_{\star}^{\text{FRI}} \leq \varepsilon_{\text{Merkle}} + \varepsilon_{\text{in}} + \varepsilon_{\text{link}}, \quad \varepsilon_{\text{glob}}^{\text{FRI}} \leq \varepsilon_{\text{Merkle}} + \varepsilon_{\text{out}}. \quad (23)$$

Here ε_{out} is “global”: it is checked once per commitment, while $\varepsilon_{\text{in}} + \varepsilon_{\text{link}}$ affects only the probability of falsely certifying an unrecoverable group.

Field size and link error. The one-link error satisfies $\varepsilon_{\text{link}} \leq D_{\text{link}}/(q - |A|)$ (Lemma 2). If q is not large enough to make this negligible, the verifier can request t independent link points to reduce the error to $(D_{\text{link}}/(q - |A|))^t$ (Section 5.5), trading a constant-factor increase in proof size for improved soundness.

7.3 Batching and aggregation

Multi-point openings within a group. When $U \subseteq A_j$ contains multiple sampled points, both instantiations can open all points with a single batched proof using vanishing polynomials: represent the opened evaluations as the remainder modulo $Z_U(X)$ and prove the quotient identity as in [4]. This reduces verification overhead and aligns with the LR-DAS interface in Section 3.2.

Batching across many groups. For KZG, random linear combination techniques can aggregate multiple opening equations into $O(1)$ pairing checks, and for FRI, standard batching reduces verifier work to a small number of LDT verifications plus Merkle checks (see [3,8] for batching patterns). These optimizations preserve the one-link condition and therefore preserve the graded-soundness accounting in Section 6.

8 Comparison and Evaluation

This section contextualizes LR-DAS among representative DAS designs and reports a compact, reproducible metric comparison. The goal is twofold: (i) clarify the trade-offs against RS-only baselines (including 2D-RS-inspired designs), and (ii) make explicit which benefits of LR-DAS (notably graded availability) are *qualitative* and therefore not fully captured by bandwidth-only metrics.

8.1 Schemes compared

We compare the following families of DAS schemes.

RS-only baselines (binary availability). We include (a) a hash/Merkle-based erasure-code commitment baseline representative of the framework in [7], and (b) FRIDA, a transparent DAS construction based on FRI/IOPP techniques [8,3]. These schemes are designed around RS-style global recovery thresholds and typically expose a binary availability signal.

2D-inspired designs (geometric organization). The blockchain industry, particularly within the Ethereum ecosystem, has actively proposed and researched 2D data availability sampling structures to enhance scalability and fault tolerance [6,11]. In line with these developments, our analysis includes 2D-structured DAS schemes based on hashes and FRI+RS, as specifically presented in the foundations and constructions of FRIDA [7,8]. These 2D organizations provide robust operational repair paths, though the availability decision typically remains a probabilistic property tied to global recovery.

MultDAS (repair-first). We include MultDAS from [5], both with and without batching. MultDAS is optimized for repair bandwidth via multiplicity-style coding and multivariate/derivative-aware commitments.

LR-DAS (this work). We include our two instantiations: LR-DAS(KZG) using pairing-based polynomial commitments [10] and LR-DAS(FRI) using a transparent commitment stack grounded in FRI [3,8]. Both use the one-point local-global link of Section 5 and batching via vanishing polynomials [4].

8.2 Common setting and metrics

Setting. Following prior evaluations (notably [5]), we normalize all schemes to a 3 MB block. As specified in Table 3, we adopt a code rate of $1/2$ for KZG-based schemes and a rate of $1/4$ for FRI-based approaches to ensure a fair comparison.

For both LR-DAS instantiations, we fix the locality parameter at $r = 512$. The number of groups ℓ is then determined for each scheme to maintain the target block size and code rates according to the design in Sections 3–5, where $n = \ell(r + \alpha)$. Furthermore, the per-query response measurements reflect the batching strategy where we specify the opening value for a single point.

Table 3. Comparison for a 3 MB block. Regarding KZG-based schemes, FuluDAS utilizes a code rate of $1/2$, and our LR-DAS (KZG) similarly adopts $1/2$ (MultDAS claims to achieve 0.53). For FRI-based approaches, FRIDA employs a rate of $1/4$, and our LR-DAS (FRI) uses $1/4$ for comparison. The implementations of LR-DAS specify the open value for a single point in the per-query response. These measurements for LR-DAS reflect the batching strategy described in Section 5.5. Both LR-DAS instantiations use locality parameter $r = 512$. In this setting, LR-DAS (KZG) has a local structure with $\ell = 123$, and LR-DAS (FRI) with $\ell = 16$.

Metric	FuluDAS ([12,5])	Hash ([7])	FRIDA ([8])	MultDAS ([5])	MultDAS (batched)	LR-DAS (KZG)	LR-DAS (FRI)
Per-query response	196.5 KB	3.47 KB	2.72 KB	5.6 KB	45 KB	0.43 KB	1.41 KB
Local repair: total bandwidth.	3 MB	3 MB	3 MB	511.9 KB	1440 KB	24.96 KB	426.56 KB
Local repair: number of sources	64	867	8192	91	32	512	512
Dispersal units	128	3,468	32,768	9,216	1,225	125,952	32,768
Commitment	1.12 KB	443.9 KB	321.6 KB	1.12 KB	9 KB	11.7 KB	6.99 MB
Transparent setup	No	Yes	Yes	No	No	No	Yes
Locality Proof	No	No	No	Yes (Repair Safety)	Yes (Repair Safety)	Yes (4)	Yes (4)
Monotone disjoint-group certification	No	No	No	No	No	Yes	Yes

Metrics. We report: (i) *Per-query response* (called “Node storage” in [5]): bytes communicated to answer a single sampled position (index bits + one symbol + opening proof/metadata), (ii) *Local repair bandwidth*: total bytes needed to repair *one local group*, (iii) *Local repair sources*: number of dispersal units contacted to repair one group (equal to locality r), (iv) *Dispersal units*: total number of symbol positions used for dispersal, (v) *Commitment size*: total verification metadata that must be distributed with the block, and (vi) whether the setup is *transparent*, (vii) *Locality Proof*: whether the scheme provides a proof to ensure the integrity and safety of the local repair process, Finally, we explicitly call out whether the scheme naturally supports *monotone disjoint-group graded certification* as in Section 3.3.

8.3 Discussion and Implications

Graded availability offers a qualitative advantage. Table 3 reports bandwidth and metadata costs, but LR-DAS’ main benefit is semantic: a verifier can certify a *fraction* of groups early (Section 3.3), enabling (i) early warning that a block is *not* progressing toward full availability, (ii) targeted re-sampling/repair focused on the remaining uncertified groups, and (iii) protocol policies that react to partial evidence (e.g., conservative relay only after a threshold fraction is certified). These behaviors are difficult to express in purely RS-threshold-based schemes without additional structure.

LR-DAS vs. RS-only baselines (Hash, FRIDA). Compared to RS-only designs, LR-DAS reduces *local repair bandwidth* by allowing recovery within a group (using only r distinct openings inside one group), while still supporting global recovery when enough groups are certified. The trade-off is that LR-DAS must incorporate locality metadata and linking proofs: its commitment sizes are larger than the smallest succinct 2D-style commitments, but substantially smaller than bandwidth-intensive RS-only transparent designs (e.g., Hash/FRIDA commitment sizes).

LR-DAS vs. MultDAS (repair-first). MultDAS provides the smallest repair bandwidth among the compared repair-capable schemes (511.9 KB without batching in Table 3), reflecting its goal of minimizing repair cost. LR-DAS instead targets a simpler univariate pipeline and graded certification. In settings where the primary objective is *repair bandwidth* at any cost, MultDAS is highly effective; in settings where progressive certification and a simpler commitment/proof stack are priorities, LR-DAS provides a streamlined alternative.

LR-DAS vs. 2D-RS layouts: why disjoint groups matter. A 2D-RS layout provides locality along rows/columns, but the corresponding recovery sets overlap heavily. As a result, defining a *monotone* graded score analogous to $\phi(S)$ is subtle: certifying a row may partially certify many columns and vice versa, and the verifier must avoid double-counting evidence. Moreover, enforcing consistency in two dimensions typically requires multivariate commitments or more complex proof composition [6]. In contrast, LR-DAS uses *disjoint* recovery groups by design, making the certified score $\phi(S)$ well-defined and monotone, and the one-link mechanism of Section 5 ties each certified group to a single global object with a minimal interface. We provide a formal justification in Appendix D.

Connecting to graded soundness. Theorem 1 demonstrates that false certification (overshoot) is bounded by a sampling term times a cryptographic term. The table-level costs in Table 3 therefore translate into an explicit design space: larger r increases the sampling budget required to certify a group but reduces the probability of accidentally sampling enough distinct points in an unrecoverable group; the link error can be made negligible by repetition or by choosing an appropriate field (Section 5.5).

Comparison of Commitment and Verification Costs. Performance measurements show that MultDAS achieves a commitment time of 44 ms and a verification time of 2.9 ms. In comparison, LR-KZG requires significantly more time for commitment at 2337.46 ms and 5.86 ms for verification. While LR-FRI offers the fastest performance with a commitment time of 16.29 ms and a verification time of 0.28 ms, it results in a larger proof size of 1.41 KB, whereas both MultDAS and LR-KZG maintain a compact proof size of 144 bytes.

9 Proof of Custody as an Application

This section describes a simple application of LR-DAS: a lightweight *proof of custody* (PoC) layer. The goal is to enforce that an entity who published a data commitment continues to possess enough information to answer availability-style queries over time. Our PoC is conceptually related to classical proofs of storage such as provable data possession (PDP) and proofs of retrievability (POR) [2,9], but here we reuse the LR-DAS opening interface rather than designing a dedicated storage authenticator.

9.1 Setting and interface

Fix the LR-DAS public parameters $(\mathbb{F}_q, A = \bigsqcup_{j=0}^{\ell-1} A_j, r, \alpha)$ and a locality-aware erasure-code commitment as in Section 4. A prover publishes an LR-DAS commitment com to an encoded object (e.g., a TB codeword) and is then periodically challenged to answer openings.

We model challenges as derived from *public randomness* that becomes available *after* com is fixed. This can be instantiated either interactively (the verifier samples fresh randomness after receiving the prover’s commitments), or non-interactively via Fiat–Shamir with domain separation, provided the transcript hashed into the challenge includes all commitments that must be fixed before the challenge (notably the quotient commitments in the one-link check of Section 5).

9.2 Protocol

We describe two complementary challenge types.

Global custody (publish-time check). After com is published, the verifier samples a fresh out-of-domain point $\rho \in \mathbb{F}_q \setminus A$ and requests a point opening of the globally committed object at ρ . In the KZG instantiation this is a standard KZG evaluation opening under the global commitment; in the transparent instantiation it corresponds to a DEEP/out-of-domain query inside the IOPP stack. This check is inexpensive (one point opening) and ensures the prover can answer an unforeseeable evaluation query tied to the commitment.

Periodic local custody (ongoing storage checks). At each epoch t , public randomness determines a set $J_t \subseteq [\ell]$ of targeted groups (e.g., $|J_t| = u$). For each $j \in J_t$, the prover returns a batched opening for the entire group A_j (or, optionally, for a randomly chosen subset $U \subseteq A_j$ of size at least r), together with the LR-DAS local-verification material for that group:

1. a proof that the opened values are consistent with a degree- $< r$ local polynomial δ_j (local membership), and
2. the one-link local-global consistency proof at a fresh out-of-domain point ρ_t as in Section 5, i.e., an authenticated check of

$$f(\rho_t) - \delta_j(\rho_t) = Z_{A_j}(\rho_t) \cdot q_j(\rho_t),$$

where f denotes the globally committed polynomial and q_j is the quotient.

The verifier accepts the epoch if all challenged groups pass verification.

Remark (what PoC guarantees in our setting). Because LR-DAS codewords are defined by a compact witness (a univariate polynomial), a prover that retains the encoding polynomial can answer custody challenges without explicitly storing all dispersed symbols. Thus, PoC here should be understood as certifying *possession of sufficient information to compute the committed codeword*, which is the natural analogue of possession in many proofs-of-storage formulations [2,9].

9.3 Soundness bound

We derive a compact bound for the common missing-groups adversary model. Let $W \subseteq A$ be the missing set and assume W is a union of entire groups: $W = \bigcup_{j \in M} A_j$ for some $M \subseteq [\ell]$ with $|M| = g$. Equivalently, a fraction $\theta = g/\ell$ of groups is completely missing.

Theorem 2 (PoC soundness from LR-DAS graded soundness). *Assume non-adaptive withholding (the set M is fixed before the epoch challenges are determined). Suppose each epoch targets u groups sampled uniformly without replacement from $[\ell]$, independently across epochs. Let $\varepsilon_\star$ and $\varepsilon_{\text{glob}}$ be the per-group and global failure parameters from Theorem 1 (instantiated as in Section 7).*

Then for any prover missing g whole groups, the probability of passing T epochs of periodic local custody satisfies

$$\Pr[\text{PoC accepts for all } T \text{ epochs}] \leq \left(\frac{\binom{\ell-g}{u}}{\binom{\ell}{u}} \right)^T + Tu \cdot \varepsilon_\star + \varepsilon_{\text{glob}}. \quad (24)$$

In particular, since $\binom{\ell-g}{u}/\binom{\ell}{u} \leq (1-\theta)^u$, we also have

$$\Pr[\text{accept}] \leq (1-\theta)^{uT} + Tu \cdot \varepsilon_\star + \varepsilon_{\text{glob}}.$$

If an additional publish-time global custody check is performed once (one out-of-domain point opening), its forging probability can be absorbed into $\varepsilon_{\text{glob}}$.

Proof (Proof sketch). In a single epoch, PoC acceptance occurs in one of two cases: (i) all targeted groups avoid the missing set M , which occurs with probability $\binom{\ell-g}{u}/\binom{\ell}{u}$, or (ii) at least one targeted group is missing. In case (ii), the prover must produce an accepting local certificate for a missing group, which requires a forged acceptance of the underlying locality-aware commitment checks and occurs with probability at most $u\varepsilon_\star$ by a union bound. Independence across epochs yields the first term raised to the power T , and adding the global failure probability $\varepsilon_{\text{glob}}$ completes the bound.

9.4 Overhead

PoC reuses LR-DAS openings and verification logic. In the KZG instantiation, each challenged group requires one batched opening under a degree- $< r$ commitment plus $O(1)$ additional point openings at ρ_t for the one-link check. In the transparent instantiation, each challenged group requires Merkle authentication and an inner low-degree/proximity proof plus the one-link check, all of which are already part of LR-DAS group certification. Therefore, PoC adds negligible new cryptographic machinery; it primarily adds a *scheduling policy* (when and how many groups to challenge).

10 Conclusion

We introduced *Locally Recoverable Data Availability Sampling* (LR-DAS), a DAS framework that exposes *graded* availability evidence by certifying disjoint local recovery groups. Our main contribution is a locality-aware erasure-code commitment that enforces *local-global consistency* via a single out-of-domain algebraic linking check, preventing splicing of locally consistent views across groups. This yields a clean verification pipeline: once a verifier obtains r distinct accepted openings in a group, it can reconstruct that entire group locally and monotonically increase its certified availability fraction.

We formalized graded soundness by cleanly separating probabilistic sampling from cryptographic commitments. Our overshoot bounds decompose false graded certification into a sampling term and a negligible cryptographic failure term, applying to both missing-groups and missing-fraction withholding patterns. We provided two concrete instantiations: a succinct pairing-based construction using KZG and a transparent construction based on Merkle commitments and FRI-style proximity tests, both compatible with standard batching techniques.

Several directions remain open: tightening parameters and improving practicality (especially link soundness under small fields or constrained proof budgets); extending LR-DAS to stronger adversarial settings (adaptive withholding, grinding, correlated failures) and end-to-end protocol analyses; and exploring locally recoverable codes optimized for bursty erasures, heterogeneous locality, or improved global distance under fixed locality.

Acknowledgements

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A Proofs of Main Theorems

A.1 Full proof of Theorem 1 (Overshoot bound)

Proof. Fix a missing set $W \subseteq A$ and recall the induced partition of groups $\text{Good}(W), \text{Bad}(W) \subseteq [\ell]$ from (16), as well as

$$\phi_{\max}(W) = \frac{|\text{Good}(W)|}{\ell}.$$

Let Q be the verifier's sampled multiset of positions, and for each group index $j \in [\ell]$ let $S_j := |\text{supp}(Q \cap A_j)|$ denote the number of *distinct* sampled points in group A_j (Section 3.5). Define the sampling event

$$E_j^{\text{sample}} := \{S_j \geq r\},$$

and let $p_j := \Pr[E_j^{\text{sample}}]$ be its probability over the verifier's sampling randomness. Let E_{glob} denote an arbitrary “global failure” event (e.g., failure of global binding/position binding, or any other global condition that is assumed by the analysis), and assume $\Pr[E_{\text{glob}}] \leq \varepsilon_{\text{glob}}$.

Let S be the set of accepted openings accumulated by the verifier and recall that the LR-DAS output is $\phi(S) = A(S)/\ell$, where $A(S)$ counts the number of groups certified from S (Section 3.3). Define the overshoot event

$$\text{Over}(W) := \{\phi(S) > \phi_{\max}(W)\}.$$

Step 1: Overshoot implies certifying a bad group. If $\phi(S) > \phi_{\max}(W)$, then $A(S) > |\text{Good}(W)|$, hence at least one certified group must lie in $\text{Bad}(W)$. Therefore,

$$\text{Over}(W) \subseteq \bigcup_{j \in \text{Bad}(W)} \text{Cert}_j, \quad (25)$$

where Cert_j is the event that the verifier certifies group A_j .

Taking probabilities, splitting off the global failure event, and using the union bound gives

$$\Pr[\text{Over}(W)] \leq \Pr[E_{\text{glob}}] + \Pr[\text{Over}(W) \wedge \neg E_{\text{glob}}] \leq \varepsilon_{\text{glob}} + \sum_{j \in \text{Bad}(W)} \Pr[\text{Cert}_j]. \quad (26)$$

Step 2: Bounding certification of a bad group. Fix any $j \in \text{Bad}(W)$. By definition of LR-DAS certification, certifying A_j requires at least r distinct accepted points in A_j . In particular, Cert_j implies E_j^{sample} . Hence,

$$\Pr[\text{Cert}_j] = \Pr[E_j^{\text{sample}}] \cdot \Pr[\text{Cert}_j \mid E_j^{\text{sample}}] = p_j \cdot \Pr[\text{Cert}_j \mid E_j^{\text{sample}}]. \quad (27)$$

We now justify the conditional bound. Since $j \in \text{Bad}(W)$, we have

$$\text{avail}(j; W) = |A_j \setminus W| < r.$$

Therefore, on the event E_j^{sample} (which gives r *distinct* sampled points in A_j), the verifier's distinct queried set $U_j := \text{supp}(Q \cap A_j)$ must satisfy $U_j \cap W \neq \emptyset$. Equivalently, any attempted certification of A_j necessarily involves producing at least one accepting opening for a missing coordinate. By the assumption of Theorem 1, for every $j \in \text{Bad}(W)$ the probability that the verifier accepts (certifies) group A_j , conditioned on $S_j \geq r$, is at most $\varepsilon_\star$. That is,

$$\Pr[\text{Cert}_j \mid E_j^{\text{sample}}] \leq \varepsilon_\star.$$

Substituting into (27) yields

$$\Pr[\text{Cert}_j] \leq p_j \varepsilon_\star. \quad (28)$$

Step 3: Conclude. Plugging (28) into (26) gives

$$\Pr[\phi(S) > \phi_{\max}(W)] = \Pr[\text{Over}(W)] \leq \sum_{j \in \text{Bad}(W)} p_j \varepsilon_\star + \varepsilon_{\text{glob}},$$

which is exactly the bound claimed in Theorem 1.

A.2 Full proof of Theorem 2 (PoC soundness)

Proof. We use the notation of Section 9. Let $[\ell] = \{0, 1, \dots, \ell - 1\}$ index the ℓ disjoint recovery groups. Assume the missing set is a union of whole groups:

$$W = \bigcup_{j \in M} A_j \quad \text{for some } M \subseteq [\ell], \quad |M| = g,$$

and define $\theta := g/\ell$. Let E_{glob} be a global failure event with $\Pr[E_{\text{glob}}] \leq \varepsilon_{\text{glob}}$.

In each epoch $t \in \{1, \dots, T\}$, the verifier samples a challenge set $J_t \subseteq [\ell]$ of size u uniformly without replacement, and the J_t are sampled independently across epochs. The prover answers each challenged group, and the verifier accepts epoch t if all checks for all $j \in J_t$ pass.

Let Acc_T be the event that the verifier accepts *all* T epochs.

Step 1: Probability of avoiding missing groups in all epochs. For a single epoch t , define the event

$$B_t := \{J_t \cap M = \emptyset\},$$

i.e., all challenged groups are among the $\ell - g$ non-missing groups. Since J_t is uniform among all $\binom{\ell}{u}$ size- u subsets, we have

$$\Pr[B_t] = \frac{\binom{\ell-g}{u}}{\binom{\ell}{u}}. \quad (29)$$

Because epochs are sampled independently,

$$\Pr\left[\bigwedge_{t=1}^T B_t\right] = \left(\frac{\binom{\ell-g}{u}}{\binom{\ell}{u}}\right)^T. \quad (30)$$

Step 2: If a missing group is ever challenged and all epochs accept, then a forgery occurred. If Acc_T holds but $\bigwedge_{t=1}^T B_t$ fails, then there exists an epoch t^* such that $J_{t^*} \cap M \neq \emptyset$. Since Acc_T implies the verifier accepted epoch t^* , the verifier must have accepted at least one challenged missing group in that epoch.

To formalize a union bound over at most Tu group-check attempts, fix an arbitrary deterministic ordering of each subset $J_t = \{J_{t,1}, \dots, J_{t,u}\}$ (e.g., ascending order). For each pair (t, i) define the event

$$F_{t,i} := \{J_{t,i} \in M \wedge \text{the verifier accepts the proof(s) for group } A_{J_{t,i}} \text{ in epoch } t\}.$$

Then

$$\text{Acc}_T \wedge \neg \bigwedge_{t=1}^T B_t \subseteq \bigcup_{t=1}^T \bigcup_{i=1}^u F_{t,i}. \quad (31)$$

Step 3: Bound acceptance probability. We split on the global failure event:

$$\Pr[\text{Acc}_T] \leq \Pr[E_{\text{glob}}] + \Pr[\text{Acc}_T \wedge \neg E_{\text{glob}}] \leq \varepsilon_{\text{glob}} + \Pr[\text{Acc}_T \wedge \neg E_{\text{glob}}].$$

Next, using (31) and the union bound,

$$\begin{aligned} \Pr[\text{Acc}_T \wedge \neg E_{\text{glob}}] &\leq \Pr\left[\bigwedge_{t=1}^T B_t\right] + \Pr\left[\left(\bigcup_{t=1}^T \bigcup_{i=1}^u F_{t,i}\right) \wedge \neg E_{\text{glob}}\right] \\ &\leq \left(\frac{\binom{\ell-g}{u}}{\binom{\ell}{u}}\right)^T + \sum_{t=1}^T \sum_{i=1}^u \Pr[F_{t,i}]. \end{aligned} \quad (32)$$

It remains to bound $\Pr[F_{t,i}]$. If $J_{t,i} \notin M$ then $F_{t,i}$ is false. If $J_{t,i} \in M$, then the challenged group is wholly missing, and in our PoC protocol the verifier requests an opening that includes at least r distinct points in that group (e.g., opening the full group or any subset of size $\geq r$). Thus, any acceptance for such a missing group corresponds to the same type of per-group “false certification” event captured by the parameter $\varepsilon_\star$ from Theorem 1. Therefore, for every (t, i) ,

$$\Pr[F_{t,i}] \leq \varepsilon_\star.$$

Substituting this into (32) yields

$$\Pr[\text{Acc}_T] \leq \left(\frac{\binom{\ell-g}{u}}{\binom{\ell}{u}}\right)^T + Tu \cdot \varepsilon_\star + \varepsilon_{\text{glob}},$$

which proves the first inequality in Theorem 2.

Step 4: Simplified bound via $\theta = g/\ell$. We have the standard bound

$$\frac{\binom{\ell-g}{u}}{\binom{\ell}{u}} = \prod_{i=0}^{u-1} \frac{\ell-g-i}{\ell-i} \leq \left(\frac{\ell-g}{\ell}\right)^u = (1-\theta)^u.$$

Raising both sides to the power T gives

$$\left(\frac{\binom{\ell-g}{u}}{\binom{\ell}{u}} \right)^T \leq (1-\theta)^{uT},$$

which proves the simplified inequality in Theorem 2.

B Security of Instantiations

This appendix shows that the instantiations in Section 7 satisfy the security requirements in Definitions 2–5 (position binding, local binding, local–global consistency, and reconstruction binding to the designated code C). We keep the arguments modular by isolating each source of failure.

Notation for failure probabilities. We use the following shorthand (all as functions of the security parameter, omitted for readability):

- ε_{KZG} : an upper bound on the probability of breaking KZG evaluation binding for any commitment used by the scheme (global, local, or quotient commitments) [10].
- $\varepsilon_{\text{Merkle}}$: an upper bound on forging two different openings to the same Merkle root (collision/second-preimage failure for the underlying hash; concretely, the probability that two distinct leaves admit valid authentication paths to the same root).
- ε_{in} : soundness error of the *inner* (per-row) low-degree/proximity proof used to justify that a row is consistent with a degree- $< r$ polynomial.
- ε_{out} : soundness error of the *outer* proof Π_{out} (defined precisely in Section B.2) that binds the commitment to a single global TB codeword (or, more precisely, to the code C specified there).
- $\varepsilon_{\text{link}}$: the one-link algebraic error from Lemma 2 (Section 5), e.g., $\varepsilon_{\text{link}} \leq D_{\text{link}}/(q - |A|)$, or amplified by t repetitions as in Section 5.5.

B.1 KZG-LAECC satisfies Definitions 2–5

We consider the KZG-based instantiation from Section 7.1. Recall that the verifier checks: (i) a batched KZG opening of the local polynomial δ_j on $U \subseteq A_j$, and (ii) the one-link identity at a transcript-derived out-of-domain challenge $\rho \notin A$ using a committed quotient q_j .

Code used for reconstruction binding. The KZG instantiation in Section 7.1 targets the systematic regime $k = \ell r$ and assembles

$$f(X) = \sum_{j=0}^{\ell-1} s_j(X) \delta_j(X),$$

where the selectors $s_j(X)$ are public and satisfy $s_j|_{A_j} \equiv 1$ and $s_j|_{A_i} \equiv 0$ for $i \neq j$. In this regime, any choice of degree- $< r$ local polynomials $\{\delta_j\}_{j=0}^{\ell-1}$ defines a unique codeword $c \in C$ by $c(a) = f(a)$ for all $a \in A$. Equivalently, C can be viewed as the set of vectors whose restriction to each A_j is the evaluation of a degree- $< r$ polynomial.

Lemma 4 (KZG-LAECC position binding (Definition 2)). *Assuming KZG evaluation binding with failure probability at most ε_{KZG} , the KZG-LAECC instantiation is ε_{KZG} -position binding.*

Proof. Suppose an adversary produces two accepting openings for the same commitment com and the same point $a \in A$ but with two different values. Any accepting opening at a must verify either against the global KZG commitment C_{glob} (a point opening of f), or against the local commitment $C_{\text{loc}}^{(j)}$ for the unique group j with $a \in A_j$ (a point opening of δ_j , possibly as part of a batched opening). In either case, two different accepted values for the same KZG commitment at the same evaluation point contradict KZG evaluation binding, except with probability at most ε_{KZG} .

Lemma 5 (KZG-LAECC local binding (Definition 3)). *Assuming KZG evaluation binding with failure probability at most ε_{KZG} , the KZG-LAECC instantiation is ε_{KZG} -local binding.*

Proof. Fix a group index j . The commitment com contains a KZG commitment $C_{\text{loc}}^{(j)}$ to a polynomial δ_j of degree at most $r - 1$ (since the SRS supports degree bound $r - 1$). Any accepting local opening on any $U \subseteq A_j$ must be a valid KZG multi-point opening of this *same* committed polynomial. Therefore, for any collection of accepting local openings for group A_j , all opened values must equal $\delta_j(\cdot)$ on their respective points, except with probability at most ε_{KZG} . Since δ_j has degree $< r$, the union of these opened values is consistent with the degree- $< r$ polynomial δ_j . Thus, producing two accepting local openings that are inconsistent with *any* degree- $< r$ polynomial implies a KZG opening forgery, contradicting evaluation binding except with probability ε_{KZG} .

Lemma 6 (KZG-LAECC local-global consistency (Definition 4)). *Assuming KZG evaluation binding with failure probability at most ε_{KZG} and the one-link check with algebraic error at most $\varepsilon_{\text{link}}$, the KZG-LAECC instantiation satisfies Definition 4 with failure probability at most $\varepsilon_{\text{KZG}} + \varepsilon_{\text{link}}$.*

Proof. Fix a group A_j and a point $a \in A_j$. Assume the verifier accepts a local opening that yields value $v = \delta_j(a)$ and also accepts a global point opening that yields value $y = f(a)$, with $v \neq y$. By Lemma 4, except with probability ε_{KZG} the two accepted openings bind v and y to the unique committed polynomials δ_j and f .

Now consider the link proof for group A_j included in verification. The verifier checks (at transcript-derived $\rho \notin A$) that

$$f(\rho) - \delta_j(\rho) = Z_{A_j}(\rho) q_j(\rho)$$

for a committed quotient q_j , with commitments to all three polynomials fixed before ρ is derived. By Lemma 2 (Section 5), except with probability $\varepsilon_{\text{link}}$ this implies the polynomial identity $f(X) - \delta_j(X) = Z_{A_j}(X) q_j(X)$, and hence $f|_{A_j} = \delta_j|_{A_j}$. In particular, $f(a) = \delta_j(a)$, contradicting $v \neq y$. Therefore, the event $v \neq y$ can occur only if either KZG binding fails or the link check errs, with total probability at most $\varepsilon_{\text{KZG}} + \varepsilon_{\text{link}}$.

Lemma 7 (KZG-LAECC reconstruction binding to C (Definition 5)).

Assuming KZG evaluation binding with failure probability at most ε_{KZG} and link error at most $\varepsilon_{\text{link}}$, the KZG-LAECC instantiation is $(\varepsilon_{\text{KZG}} + \varepsilon_{\text{link}})$ -reconstruction binding to the code C described above.

Proof. We describe an extractor E for Definition 5. Given (ck, com) with $\text{com} = (C_{\text{glob}}, \{C_{\text{loc}}^{(j)}\}_{j \in [\ell]})$, the extractor interacts with the adversary as a verifier and, for each group $j \in [\ell]$, requests a local opening on a fixed public set $U_j \subseteq A_j$ of size r . If the adversary produces an accepting transcript, the extractor interpolates the unique degree- $< r$ polynomial δ_j^* from the r opened pairs in U_j . After doing this for all groups for which the adversary provides accepting transcripts, the extractor defines

$$f^*(X) := \sum_{j=0}^{\ell-1} s_j(X) \delta_j^*(X)$$

and outputs the codeword $c^* \in C$ given by $c^*(a) = f^*(a)$ for all $a \in A$.

We argue that, except with probability at most $\varepsilon_{\text{KZG}} + \varepsilon_{\text{link}}$, every *accepted* opening that the adversary can produce (under **VerifyLocal** and **VerifyPoint** when applicable) reveals $c^*(a)$. Fix any accepted opening at a point $a \in A_j$. By Lemma 5, except with probability ε_{KZG} the opened value equals the committed polynomial δ_j evaluated at a . By Lemma 6, except with probability $\varepsilon_{\text{KZG}} + \varepsilon_{\text{link}}$ we also have $\delta_j(a) = f(a)$ for the committed global polynomial f . Finally, by construction of f^* and the selector property $s_j|_{A_j} \equiv 1$, we have $f^*(a) = \delta_j^*(a)$, and δ_j^* agrees with δ_j on the r extracted points. Since both are degree- $< r$, they coincide on all points where the adversary can produce accepting openings without violating KZG evaluation binding. Therefore, every accepted opening reveals $f^*(a) = c^*(a)$ except with total failure probability at most $\varepsilon_{\text{KZG}} + \varepsilon_{\text{link}}$.

B.2 FRI-LAECC satisfies Definitions 2–5

We now analyze the transparent instantiation from Section 7.2. The commitment contains a Merkle root M_D of an evaluation table D over the domain $A = \bigsqcup_{j=0}^{\ell-1} A_j$, and an outer proof Π_{out} . For each certified group A_j , verification additionally uses: (i) Merkle authentication paths for opened coordinates, (ii) an inner proof $\Pi_{\text{in}}^{(j)}$ that the row is degree- $< r$, and (iii) the one-link check at an out-of-domain point $\rho \notin A$.

Defining $\mathcal{R}_{\text{out}}$ and Π_{out} This subsection fixes the exact meaning of “outer binding to a single TB codeword” used in Section 7.2.

Public group labels. Let $g(X)$ be the public good polynomial used to define the disjoint groups, so g is constant on each A_j . Write $\gamma_j := g(a)$ for any $a \in A_j$, and let $\Gamma := \{\gamma_0, \dots, \gamma_{\ell-1}\}$.

TB structure as a functional constraint. Recall the standard TB encoding view (Section 2): there exist coefficient polynomials $p_0(T), \dots, p_{r-1}(T) \in \mathbb{F}_q[T]$ with prescribed degree bounds such that the global codeword is described by

$$f(X) = \sum_{i=0}^{r-1} p_i(g(X)) X^i. \quad (33)$$

When restricted to A_j , the values $p_i(g(X))$ are constants $p_i(\gamma_j)$, so the restriction $f|_{A_j}$ is a degree- $< r$ polynomial in X , matching local recoverability.

Outer relation $\mathcal{R}_{\text{out}}$. Let $D : A \rightarrow \mathbb{F}_q$ denote the committed table values (i.e., $D(a)$ is the Merkle-committed value at position $a \in A$). Let κ_i denote the public degree bounds for the coefficient polynomials p_i (as fixed by the TB parameters in Section 2).

We define the outer relation $\mathcal{R}_{\text{out}}$ as follows: an instance of the relation consists of $(\text{ck}, M_D, \{M_{P,i}\}_{i=0}^{r-1})$, where each $M_{P,i}$ is a Merkle root committing to a table $P_i : \Gamma \rightarrow \mathbb{F}_q$ intended to encode evaluations of p_i on Γ . A witness consists of polynomials $\{p_i(T)\}_{i=0}^{r-1}$ and a polynomial $f(X)$. The relation $\mathcal{R}_{\text{out}}(\text{ck}, M_D, \{M_{P,i}\}; f, \{p_i\}) = 1$ holds iff:

1. $\deg(f) \leq D$ and for every $a \in A$, the committed value satisfies $D(a) = f(a)$;
2. for each $i \in \{0, \dots, r-1\}$, $\deg(p_i) < \kappa_i$ and for every $\gamma \in \Gamma$, $P_i(\gamma) = p_i(\gamma)$;
3. for every $a \in A$, the functional identity holds:

$$D(a) = \sum_{i=0}^{r-1} p_i(g(a)) a^i. \quad (34)$$

If the relation holds, then the committed table D is exactly the evaluation vector of a single TB codeword $c(a) = f(a)$ over A .

Outer proof Π_{out} . We instantiate Π_{out} by a standard transparent IOPP/FRI stack that proves (i) low degree of the relevant tables and (ii) the identity (34), with all challenges derived from a domain-separated Fiat-Shamir transcript. Concretely, the proof includes:

- a (transparent) low-degree proof that the table committed by M_D is the evaluation of a degree- $\leq D$ polynomial f ;
- for each i , a low-degree proof that the table committed by $M_{P,i}$ is the evaluation of a degree- $< \kappa_i$ polynomial p_i ;
- an identity test (using random challenges and low-degree checks) enforcing (34).

We assume the resulting system has soundness error at most ε_{out} : if no witness exists for $\mathcal{R}_{\text{out}}$, then the verifier accepts Π_{out} with probability at most ε_{out} . This assumption matches the standard way FRI-based IOPPs yield code commitments to structured RS-derived relations [3,8].

Security reductions for FRI-LAECC

Lemma 8 (FRI-LAECC position binding (Definition 2)). *Assuming Merkle position binding with failure probability at most $\varepsilon_{\text{Merkle}}$, the FRI-LAECC instantiation is $\varepsilon_{\text{Merkle}}$ -position binding.*

Proof. Any accepted opening at a point $a \in A$ must include a Merkle authentication path to the root M_D . If an adversary produces two accepting openings for the same a but with two different values, then there exist two different leaf values with valid authentication paths to the same Merkle root, contradicting Merkle binding except with probability $\varepsilon_{\text{Merkle}}$.

Lemma 9 (FRI-LAECC local binding (Definition 3)). *Assuming Merkle binding with failure probability at most $\varepsilon_{\text{Merkle}}$ and inner proof soundness error at most ε_{in} , the FRI-LAECC instantiation is $(\varepsilon_{\text{Merkle}} + \varepsilon_{\text{in}})$ -local binding.*

Proof. Fix a group index j . Any accepted local opening on $U \subseteq A_j$ reveals values $D(a)$ for $a \in U$ together with valid Merkle paths to M_D . Thus, except with probability $\varepsilon_{\text{Merkle}}$, all accepted local openings for group A_j reveal values from the same committed row $D|_{A_j}$.

The verifier additionally accepts an inner proof $\Pi_{\text{in}}^{(j)}$ asserting that the full row $D|_{A_j}$ is consistent with a degree- $< r$ polynomial δ_j on A_j . If the collection of accepted openings in A_j is inconsistent with *any* degree- $< r$ polynomial, then either: (i) some opened value is not the committed Merkle value (a Merkle binding failure), or (ii) the inner proof accepted even though $D|_{A_j}$ is not degree- $< r$ consistent (an inner-soundness failure). This yields a total failure probability at most $\varepsilon_{\text{Merkle}} + \varepsilon_{\text{in}}$.

Lemma 10 (FRI-LAECC local-global consistency (Definition 4)). *Assuming Merkle binding with failure probability at most $\varepsilon_{\text{Merkle}}$, the FRI-LAECC instantiation satisfies Definition 4 with failure probability at most $\varepsilon_{\text{Merkle}}$.*

Proof. In the transparent instantiation, a “global” opening at a point $a \in A$ is simply the Merkle-authenticated value $D(a)$. A “local” opening at a point $a \in A_j$ is also the Merkle-authenticated value $D(a)$ (plus additional structure proofs). Thus, producing an accepting local opening at a that is inconsistent with an accepting global opening at the same a requires two different authenticated values for the same Merkle leaf, contradicting Merkle binding except with probability $\varepsilon_{\text{Merkle}}$.

Lemma 11 (FRI-LAECC reconstruction binding to C (Definition 5)). *Assume Merkle binding with failure probability at most $\varepsilon_{\text{Merkle}}$, outer proof soundness error at most ε_{out} , inner proof soundness error at most ε_{in} , and one-link error at most $\varepsilon_{\text{link}}$. Then the FRI-LAECC instantiation is $(\varepsilon_{\text{Merkle}} + \varepsilon_{\text{out}} + \varepsilon_{\text{in}} + \varepsilon_{\text{link}})$ -reconstruction binding to the TB code C induced by (33).*

Proof. We describe an extractor E as required by Definition 5. Given (ck, com) with $\text{com} = (M_D, \Pi_{\text{out}})$ (and with the auxiliary Merkle roots $\{M_{P,i}\}$ included

in Π_{out}), the extractor first runs the verifier for Π_{out} . If Π_{out} is rejected, the extractor outputs an arbitrary codeword in C .

Otherwise, except with probability ε_{out} , there exists a witness $(f, \{p_i\})$ satisfying the relation $\mathcal{R}_{\text{out}}$ from Section B.2. Conditioned on this event, define $c^* \in C$ by $c^*(a) = f(a)$ for all $a \in A$.

We argue that every accepted opening must reveal $c^*(a)$ except with probability $\varepsilon_{\text{Merkle}} + \varepsilon_{\text{in}} + \varepsilon_{\text{link}}$. Fix any accepting transcript for an opening at a point $a \in A$ (either via `VerifyPoint` or as part of `VerifyLocal`). By Merkle verification, except with probability $\varepsilon_{\text{Merkle}}$, the opened value equals the committed value $D(a)$.

Next, because Π_{out} is accepted and $\mathcal{R}_{\text{out}}$ holds for the extracted witness, we have $D(a) = f(a) = c^*(a)$ for all $a \in A$ by item (1) in the definition of $\mathcal{R}_{\text{out}}$. Therefore, any accepted opening at a domain point a reveals $c^*(a)$ except with probability $\varepsilon_{\text{Merkle}} + \varepsilon_{\text{out}}$.

Finally, Definition 5 also quantifies over accepting *local* transcripts in which the verifier may reconstruct additional symbols in a group using the degree- $< r$ local polynomial. In our scheme, this reconstruction is justified by: (i) the inner proof $\Pi_{\text{in}}^{(j)}$ (soundness error ε_{in}) which ensures that the committed row $D|_{A_j}$ matches a degree- $< r$ polynomial δ_j on A_j , and (ii) the one-link check (soundness error $\varepsilon_{\text{link}}$) which ensures that δ_j equals the restriction of the *same* global polynomial f (as in Section 5). Therefore, except with probability $\varepsilon_{\text{in}} + \varepsilon_{\text{link}}$, any value reconstructed from δ_j for group A_j equals $f(\cdot) = c^*(\cdot)$ on that group. Combining all failure events by a union bound yields the stated error $\varepsilon_{\text{Merkle}} + \varepsilon_{\text{out}} + \varepsilon_{\text{in}} + \varepsilon_{\text{link}}$.

C Minimality of the Linking Mechanism

This appendix explains why LR-DAS needs an explicit *link* between the globally committed object and the per-group local witness used for certification and reconstruction. Informally, without such a link (or an equivalent amount of additional checking), a prover can provide a locally consistent degree- $< r$ witness on the sampled points while making the *reconstructed* (unopened) symbols in the group disagree with the globally committed codeword. The one-link mechanism of Section 5 is the minimal algebraic linkage that rules out this splicing behavior with only constant overhead per certified group.

C.1 A splicing attack without a link

Recall that certifying a group A_j uses a degree- $< r$ polynomial δ_j : after r distinct verified openings in A_j , the verifier interpolates δ_j and reconstructs all $r + \alpha$ symbols in A_j by evaluating δ_j on all points of A_j (Section 3.3). For this to be meaningful, the reconstructed values must agree with the *globally committed* codeword.

The following simple fact shows that local agreement on the sampled points does not imply agreement on the whole group.

Lemma 12 (Agreement on r points does not pin down a restriction).

Fix any set $U \subseteq \mathbb{F}_q$ of size $|U| = r$ and any polynomial $\delta(X) \in \mathbb{F}_q[X]$ with $\deg(\delta) < r$. For any $c \in \mathbb{F}_q^\times$, the polynomial

$$f(X) := \delta(X) + c \cdot Z_U(X)$$

satisfies $f(u) = \delta(u)$ for all $u \in U$, but $f(x) \neq \delta(x)$ for every $x \notin U$ such that $Z_U(x) \neq 0$. Moreover, $\deg(f) = \max\{\deg(\delta), r\} = r$.

Proof. Since $Z_U(u) = 0$ for all $u \in U$, we have $f(u) = \delta(u)$ on U . For any $x \notin U$ with $Z_U(x) \neq 0$, we have $f(x) - \delta(x) = c \cdot Z_U(x) \neq 0$. The degree statement follows because Z_U has degree r and $c \neq 0$.

Proposition 1 (No-link splicing: reconstructed symbols can disagree with the commitment). Fix a group A_j and let $U \subseteq A_j$ be any set of r distinct points. Consider any certification procedure that: (i) verifies openings only on U , (ii) checks that the opened values in U are consistent with some degree- $< r$ polynomial δ_j , and (iii) reconstructs the remaining points in $A_j \setminus U$ using δ_j , but does not additionally enforce that the reconstructed values agree with the globally committed object on all of A_j .

Then there exists a prover strategy that passes all checks on U yet causes the verifier to reconstruct at least one point $b \in A_j \setminus U$ to a value that is inconsistent with the globally committed codeword.

Proof. Pick any degree- $< r$ polynomial $\delta_j(X)$ and define $v_u := \delta_j(u)$ for all $u \in U$. Let $b \in A_j \setminus U$ be any point not in U (such a b exists since $|A_j| = r + \alpha > r$), so $Z_U(b) \neq 0$. Define a global polynomial

$$f(X) := \delta_j(X) + c \cdot Z_U(X)$$

for some $c \in \mathbb{F}_q^\times$ chosen so that $f(b) \neq \delta_j(b)$ (any nonzero c suffices by Lemma 12). The prover commits globally to f and locally uses δ_j as the witness for the group A_j .

On the sampled set U , the prover opens values $v_u = \delta_j(u)$. All checks on U pass by construction: the values are consistent with a degree- $< r$ polynomial (δ_j), and they also agree with the global commitment on U because $f(u) = \delta_j(u)$ for all $u \in U$. However, when the verifier reconstructs the symbol at b using δ_j , it obtains $\delta_j(b)$, whereas the globally committed codeword value at b is $f(b) \neq \delta_j(b)$. Thus the reconstructed symbol disagrees with the commitment while all checks on U succeeded.

Concrete toy example. Let $r = 2$, $\alpha = 1$, and $A_j = \{0, 1, 2\} \subseteq \mathbb{F}_q$. Take $\delta_j(X) = X$ and $U = \{0, 1\}$. Then $Z_U(X) = X(X - 1)$ and $f(X) = \delta_j(X) + Z_U(X) = X^2$. The verifier sees values $f(0) = 0$, $f(1) = 1$ (consistent with degree- < 2 polynomial $\delta_j(X) = X$), and reconstructs the missing value as $\delta_j(2) = 2$, but the committed global value is $f(2) = 4$.

C.2 Necessity of Out-of-Domain Checks

The one-link mechanism in Section 5 enforces the divisibility identity

$$f(X) - \delta_j(X) = Z_{A_j}(X) \cdot q_j(X)$$

by checking it at a random point $\rho \in \mathbb{F}_q \setminus A$. The need for an out-of-domain point is inherent.

Lemma 13 (Checking the divisibility identity inside A_j is vacuous). *Let $h_j(X) := Z_{A_j}(X)$. For any polynomials $f, \delta_j, q_j \in \mathbb{F}_q[X]$ and any $a \in A_j$, the equality*

$$f(a) - \delta_j(a) = h_j(a) \cdot q_j(a)$$

holds if and only if $f(a) = \delta_j(a)$. In particular, evaluating the identity only at points of A_j cannot certify divisibility by h_j ; it only certifies pointwise agreement at those evaluated points.

Proof. Since $h_j(a) = 0$ for all $a \in A_j$, the right-hand side is always 0. Thus the equality holds exactly when $f(a) - \delta_j(a) = 0$, i.e., $f(a) = \delta_j(a)$.

Lemma 13 shows that to certify a statement about *all* points in A_j (equivalently, divisibility by Z_{A_j}), one must evaluate at least one point outside A_j where Z_{A_j} is nonzero. This is exactly what the out-of-domain challenge $\rho \notin A$ provides.

C.3 Requirement for Pre-Challenge Commitment

Finally, the one-link check must be performed in a commit-before-challenge manner. Otherwise, a cheating prover can always satisfy the check at a *single* point without implying the polynomial identity.

Proposition 2 (Necessity of fixing q_j before ρ). *Fix any $f, \delta_j \in \mathbb{F}_q[X]$ and any $\rho \in \mathbb{F}_q \setminus A_j$. Then there exists a value $q \in \mathbb{F}_q$ such that*

$$f(\rho) - \delta_j(\rho) = Z_{A_j}(\rho) \cdot q.$$

Therefore, if the prover is allowed to choose $q_j(\rho)$ after learning ρ (without being bound to a committed polynomial q_j), the one-point check provides no soundness guarantee.

Proof. Since $\rho \notin A_j$, we have $Z_{A_j}(\rho) \neq 0$, and we may set $q := (f(\rho) - \delta_j(\rho)) / Z_{A_j}(\rho) \in \mathbb{F}_q$.

This is why our protocol commits to $q_j(X)$ (with an appropriate degree bound) *before* ρ is derived, and then checks the identity at ρ using authenticated point openings (Section 5.2 and Figure 2).

Summary. Proposition 1 shows that some form of local-global linking is necessary for safe group reconstruction. Lemma 13 shows that such a link cannot be certified purely from in-domain points of A_j . Proposition 2 shows that the quotient (or an equivalent witness) must be fixed before the challenge. The one-link construction satisfies all three requirements with constant overhead per certified group, and is therefore minimal in this sense.

D Marginal Gain Analysis: Disjoint vs. Overlapping Layouts

To strictly justify this design choice, we formalize the scoring difficulty in overlapping layouts. A sound availability score should be *additive*: verifying a new group should contribute a constant amount of distinct data. If groups overlap, as in 2D-RS, a single codeword symbol can satisfy multiple recovery constraints (e.g., both a row and a column). This leads to double-counting of availability evidence, where the same underlying data point inflates the score of multiple groups. Such structural ambiguity results in diminishing marginal gains, as subsequent certifications recover fewer *unique* symbols. By using disjoint groups, LR-DAS ensures a direct mapping between the certified availability score and the actual volume of recovered symbols.

Definition 9 (Marginal Availability Gain). *Let $\mathcal{G}$ be the set of currently certified groups, and $\text{Rec}(\mathcal{G})$ be the set of unique symbols recovered by them. For a new group $G \notin \mathcal{G}$, the marginal availability gain is defined as the volume of previously unknown data it provides:*

$$\Delta(G, \mathcal{G}) := |\text{Rec}(\mathcal{G} \cup \{G\})| - |\text{Rec}(\mathcal{G})| = |G \setminus \text{Rec}(\mathcal{G})|. \quad (35)$$

Lemma 14 (Monotonicity of Availability Gain). *Let $\mathcal{G}$ be a set of currently certified groups. For any group $G \notin \mathcal{G}$, the marginal availability gain $\Delta(G, \mathcal{G})$ (as defined in Definition 9) is strictly constant in LR-DAS, whereas it is state-dependent in overlapping layouts such as 2D-RS.*

Proof. The marginal availability gain represents the volume of new data symbols provided by G given the already recovered set $\text{Rec}(\mathcal{G})$.

1. **In LR-DAS:** The evaluation domain A is partitioned into pairwise disjoint groups $\{A_j\}_{j \in [\ell]}$. For any new group $G = A_k$ where $A_k \notin \mathcal{G}$, the disjoint property ensures $A_k \cap \text{Rec}(\mathcal{G}) = \emptyset$. Thus, the gain is invariant to the number of previously certified groups:

$$\Delta(A_k, \mathcal{G}) = |A_k| = r + \alpha. \quad (36)$$

Consequently, the availability score $\phi(S)$ increases monotonically and linearly with each certification.

2. **In 2D-RS:** Consider an $n \times n$ layout where groups consist of rows $\{R_i\}$ and columns $\{C_j\}$. Suppose $\mathcal{G}$ already contains k rows. For a new column $G = C_j$, each of the k rows intersects C_j at exactly one distinct codeword symbol. The marginal gain is therefore state-dependent:

$$\Delta(C_j, \mathcal{G}) = |C_j| - |C_j \cap (\cup_{R \in \mathcal{G}} R)| = n - k. \quad (37)$$

In this case, the gain decreases as k increases, leading to diminishing returns and potential double-counting of availability evidence.

E Full version of LAECC instantiations

KZG-LAEC (two-layer). Let $h_j(X) := Z_{A_j}(X)$.

Setup(1^λ):

1. Fix parameters for LRC $\mathcal{P} \leftarrow (\mathbb{F}_q, A = \bigsqcup_j^\ell A_j, r, \alpha, \ell)$.
2. Generate SRS:

$$\text{SRS}_{\text{glob}} \leftarrow \text{KZG.Setup}(1^\lambda, D), \quad \text{SRS}_{\text{loc}} \leftarrow \text{KZG.Setup}(1^\lambda, r - 1).$$

3. Return $\text{ck} \leftarrow (\mathcal{P}, \text{SRS}_{\text{glob}}, \text{SRS}_{\text{loc}})$.

Com(ck, m):

1. Encode m into local polynomials $\{\delta_j\}_{j=0}^{\ell-1}$ and assemble global $f(X)$.
2. Compute quotient polynomial $q_j(X) \leftarrow (f(X) - \delta_j(X))/h_j(X)$.
3. Compute KZG commitments:

$$\begin{aligned} C_{\text{glob}} &\leftarrow \text{KZG.Commit}(\text{SRS}_{\text{glob}}, f), \\ C_{\text{loc}}^{(j)} &\leftarrow \text{KZG.Commit}(\text{SRS}_{\text{loc}}, \delta_j) \quad \forall j, \\ C_q^{(j)} &\leftarrow \text{KZG.Commit}(\text{SRS}_{\text{glob}}, q_j) \quad \forall j. \end{aligned}$$

4. Set $\text{com} \leftarrow (C_{\text{glob}}, \{C_{\text{loc}}^{(j)}, C_q^{(j)}\}_{j \in [\ell]})$.
5. Set $\text{St} \leftarrow (\text{com}, f, \{\delta_j, q_j\}_{j \in [\ell]})$.
6. Return (com, St) .

OpenLocal($\text{ck}, \text{St}, j, U \subseteq A_j$):

1. Compute subset evaluations $V_U \leftarrow \{(a, \delta_j(a))\}_{a \in U}$.
2. Generate subset proof $\pi_U \leftarrow \text{KZG.OpenMulti}(\text{SRS}_{\text{loc}}, \delta_j, U)$.
3. Parse $\text{St} \rightarrow (\text{com}, f, \delta_j, q_j, \dots)$ and $\text{com} \rightarrow (C_{\text{glob}}, C_{\text{loc}}^{(j)}, C_q^{(j)})$.
4. Derive challenge $\rho \leftarrow \mathcal{H}(C_{\text{glob}} \| C_{\text{loc}}^{(j)} \| C_q^{(j)} \| j \| U \| \text{"link"})$.
5. Compute *out-of-domain* point evaluations $(v_f, v_{\delta_j}, v_{q_j}) \leftarrow (f(\rho), \delta_j(\rho), q_j(\rho))$.
6. Generate KZG single-point proofs for evaluations at point ρ :

$$\begin{aligned} \pi_f &\leftarrow \text{KZG.Open}(\text{SRS}_{\text{glob}}, f, \rho) \\ \pi_{\delta_j} &\leftarrow \text{KZG.Open}(\text{SRS}_{\text{loc}}, \delta_j, \rho) \\ \pi_{q_j} &\leftarrow \text{KZG.Open}(\text{SRS}_{\text{glob}}, q_j, \rho) \end{aligned}$$

7. Set $\pi \leftarrow (\pi_U, (v_f, v_{\delta_j}, v_{q_j}), (\pi_f, \pi_{\delta_j}, \pi_{q_j}))$.
8. Return (V_U, π) .

VerifyLocal($\text{ck}, \text{com}, j, U, V_U, \pi$):

1. Parse $\text{com} \rightarrow (C_{\text{glob}}, C_{\text{loc}}^{(j)}, C_q^{(j)}, \dots)$.
2. Parse $\pi \rightarrow (\pi_U, (v_f, v_{\delta_j}, v_{q_j}), (\pi_f, \pi_{\delta_j}, \pi_{q_j}))$.
3. Derive challenge $\rho \leftarrow \mathcal{H}(C_{\text{glob}} \| C_{\text{loc}}^{(j)} \| C_q^{(j)} \| j \| U \| \text{"link"})$.
4. Perform verifications:

$$\begin{aligned} \text{Check}_{\text{Subset}} &\leftarrow \text{KZG.VerifyMulti}(\text{SRS}_{\text{loc}}, C_{\text{loc}}^{(j)}, U, V_U, \pi_U) \\ \text{Check}_{\text{Link}} &\leftarrow (v_f - v_{\delta_j} \stackrel{?}{=} h_j(\rho) \cdot v_{q_j}) \\ \text{Check}_{\text{Point}} &\leftarrow \left\{ \begin{array}{l} \text{KZG.Verify}(\text{SRS}_{\text{glob}}, C_{\text{glob}}, \rho, v_f, \pi_f) \\ \wedge \text{KZG.Verify}(\text{SRS}_{\text{loc}}, C_{\text{loc}}^{(j)}, \rho, v_{\delta_j}, \pi_{\delta_j}) \\ \wedge \text{KZG.Verify}(\text{SRS}_{\text{glob}}, C_q^{(j)}, \rho, v_{q_j}, \pi_{q_j}) \end{array} \right\} \end{aligned}$$

5. Accept iff $\text{Check}_{\text{Subset}} \wedge \text{Check}_{\text{Link}} \wedge \text{Check}_{\text{Point}}$.

Fig. 5. KZG-based realization of Definition 1.

FRI-LAECC (Transparent). Let $h_j(X) := Z_{A_j}(X)$.

Setup(1^λ):

1. Fix parameters for LRC $\mathcal{P} \leftarrow (\mathbb{F}_q, A = \bigsqcup_j^\ell A_j, r, \alpha, \ell)$.
2. Select domain-separated hash functions $\mathcal{H}_{\text{Merkle}}$ and $\mathcal{H}_{\text{FS}}$.
3. Return $\text{ck} \leftarrow (\mathcal{P}, \mathcal{H}_{\text{Merkle}}, \mathcal{H}_{\text{FS}})$.

Com(ck, m):

1. Encode m into $f(X)$ and $\{\delta_j(X)\}_{j \in [\ell]}$, and compute quotient polynomials $q_j(X) \leftarrow (f(X) - \delta_j(X))/h_j(X)$.
2. Generate FRI proofs for all $\phi \in \{f, \delta_j, q_j\}$:

$$\Pi_\phi \leftarrow \text{FRI.Prove}(\phi)$$

Each Π_ϕ contains the merkle roots $\{R_{\phi,k}\}_k$ and authentication paths for each layer k . Let $R_\phi := R_{\phi,0}$ for simplicity.

3. Set $\text{com} \leftarrow (\Pi_f, \{\Pi_{\delta_j}, \Pi_{q_j}\}_{j \in [\ell]})$ and $\text{St} \leftarrow (\text{com}, f, \{\delta_j, q_j\}_{j \in [\ell]})$.
4. Return (com, St) .

OpenLocal($\text{ck}, \text{St}, j, U \subseteq A_j$):

1. Set $V_U \leftarrow \{(a, D_a)\}_{a \in U}$ and $\pi \leftarrow (v_f, v_{\delta_j}, v_{q_j}, \pi_f, \pi_{\delta_j}, \pi_{q_j})$.
2. Parse $\text{St} \rightarrow (\text{com}, \dots)$
3. Derive challenge $\rho \leftarrow \mathcal{H}_{\text{FS}}(R_f \| R_{\delta_j} \| R_{q_j} \| j \| U \| \text{"link"})$.
4. Compute one-link check evaluations $(v_f, v_{\delta_j}, v_{q_j}) \leftarrow (f(\rho), \delta_j(\rho), q_j(\rho))$.
5. Extract paths relative to R_ϕ in com :

$$\pi_f \leftarrow \text{Merkle.GetPath}(R_f, U),$$

$$\pi_{\delta_j} \leftarrow \text{Merkle.GetPath}(R_{\delta_j}, U),$$

$$\pi_{q_j} \leftarrow \text{Merkle.GetPath}(R_{q_j}, U).$$

6. Return (V_U, π) .

VerifyLocal($\text{ck}, \text{com}, j, U, V_U, \pi$):

1. Parse $\pi \rightarrow (v_f, v_{\delta_j}, v_{q_j}, \pi_f, \pi_{\delta_j}, \pi_{q_j})$.
2. Derive challenge $\rho \leftarrow \mathcal{H}_{\text{FS}}(R_f \| R_{\delta_j} \| R_{q_j} \| j \| U \| \text{"link"})$.
3. Verify Merkle paths, Link consistency, and Low-Degree properties:

$$\text{Check}_{\text{Merkle}} \leftarrow \bigwedge_{\phi \in \{f, \delta_j, q_j\}} \text{Merkle.Verify}(R_\phi, U, V_U, \pi_\phi)$$

$$\text{Check}_{\text{Link}} \leftarrow (v_f - v_{\delta_j} \stackrel{?}{=} h_j(\rho) \cdot v_{q_j})$$

$$\text{Check}_{\text{FRI}} \leftarrow \bigwedge_{\phi \in \{f, \delta_j, q_j\}} \text{FRI.Verify}(\Pi_\phi)$$

4. Accept iff $\text{Check}_{\text{Merkle}} \wedge \text{Check}_{\text{Link}} \wedge \text{Check}_{\text{FRI}}$.

Fig. 6. (Transparent) FRI-based realization of Definition 1.